



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

INSTITUTE OF GEOMATICS, GIS & REMOTE SENSING (IGGRoS)

ASSESSING THE IMPACT OF LAND USE AND LAND COVER CHANGES ON
HYDROLOGICAL PROCESSES IN THE THWAKE RIVER BASIN USING GIS-BASED
MODELS AND REMOTE SENSING TECHNIQUES

MAWEU MWATU

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Assessing The Impact of Land Use and Land Cover Changes on Hydrological Processes in
The Thwake River Basin Using GIS-Based Models and Remote Sensing Techniques

Maweu Mwatu

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Dedan Kimathi University of Technology, Nyeri, Kenya.

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DECLARATION

This is to confirm that I, Maweu Mwatu, whose registration number and signature appear below, undertook this project as my final year project. I also confirm that this is my original work and has not been presented in this or any other university for examination or for any other purposes.

Signature.....

Date.....

Maweu Mwatu

E032-01-1516/2019

CERTIFICATION

I confirm that the researcher under my supervision prepared the work presented in this research report. I hereby confirm the handing in of the final project report.

Signature.....



Date.....

13 Feb 2023

Supervisor

Prof. Moses Murimi Ngigi

Dedan Kimathi University of Technology

DEDICATION

I dedicate this work to my parents Mr. and Mrs. Mwatu, my aunt Florence Kaluki Maweu and my siblings Ann, Peter, and Abigail.

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First and foremost, praises and thanks to God, the Almighty, for His showers of blessings throughout my research work to complete the project successfully.

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ABSTRACT

The Thwake River basin has experienced a marked decline in rainfall over the years. The basin is geographically located in a semi-arid region where water availability for both domestic and industrial purposes is scarce. This has been coupled with an increase in population according to the 2019 population census. Moreover, land use dynamics have been known to cause considerable modifications to the environment, especially along river basins. Climate change, a natural process, affects the atmosphere's global circulation system which reflects fluctuations in flooding, drought occurrences, surface water levels, river volumes, surface water quantities, precipitations, and water vapor. The objectives of this study were: To demonstrate the capability of GIS techniques and GIS-based modelling in river basin assessment, management, and mapping, in Thwake River Basin, to perform modelling using Soil and water assessment tools, to analyse the land cover and land use changes in Thwake River Basin and to determine the positive and negative impacts of land use and land cover on the Thwake River Basin. The GIS-based modelling was carried out using the Soil and Water Assessment Tool (SWAT) which uses classified Land Use and Land Cover (LULC), soil data, and Digital Elevation Model raster datasets as inputs. The precipitation and weather time series are simulated within the model. The SWAT model was run for 10 years with 5 warm-up years. The streams flow in and flow-out results in the SWAT simulation database were visualized using graphs in Microsoft Excel. Land Use and Land Cover trends were classified from 2013 to 2022 with six classes which were: Forest, Water, Build Up, Agriculture, and Bare land. The classification accuracies were determined for each year and analysed in Microsoft Excel. The datasets used were: Digital soil maps from FAO, Sentinel imagery, and Landsat Satellite images. The results obtained were: Streams water Flow in and Flow out in the River Basin, land use, and land cover changes in the river basin. The amount of water inflow and outflowing the river basin was found to have been increasing over the years with a drastic increase in the improvement in agriculture and forest growth. From the results mentioned above, it is evident that efforts in river basin management be improved and monitored over long periods of time and simulated hence becoming effective for the management of river basins through the use of GIS-based models.

Keywords: river basin, management, GIS-based models, SWAT, climate

Table of Contents

DEDICATION.....	iii
ACKNOWLEDGMENT	iv
ABSTRACT	v
LIST OF FIGURES	viii
ABBREVIATIONS AND ACRONYMS	x
CHAPTER 1: INTRODUCTION.....	1
1.1 Background Of the study	1
1.2 Problem Statement.....	1
1.3 Justification for the study	2
1.4 Objectives.	3
1.4.1 Overall Objective.....	3
1.4.2 Specific Objectives.	3
1.5 Research Questions.....	3
1.6 Scope of work.	3
CHAPTER 2: LITERATURE REVIEW.....	4
2.1 Introduction.....	4
2.2 Overview of GIS based modelling and Remote Sensing application in river basin management.	5
2.3 Types of Basins.....	5
2.3.1 River Drainage Basin	5
2.3.2 Structural Basins	6
2.3.3 Ocean Basins.....	6
2.4 The Integration of Aquifers and Surface Water.....	7
2.5 Water table.....	8
2.6 Climate and Climate Change in Kenya.....	9
2.7 Impacts of Climate Change.....	9
2.7.1 Impact on the Natural Environment.....	9
2.7.2 Impact on Forested Watersheds	10
2.7.3 Impact on River Drainage Basins	10
2.8 River Basin Management Projects in Thwake River Basin	11
2.8.1 Thwake Dam Water Project.....	11
2.9 The Application of GIS-Based Models and RS in River Basin Management	11
2.9.1 Using GIS-Based Models to Manage River Basins.	11
2.10 The Soil and Water Assessment Tool	12
2.11 Critique of Previous Research Work	13
2.12 Research Gaps.....	13

CHAPTER 3: MATERIALS AND METHODS	14
3.1 Introduction.....	14
3.2 The Study Area	14
3.4 Data Analysis	21
3.4.1 Data Acquisition and Preparation.	21
3.4.2 Classification and Change Detection	21
3.4.3 Accuracy Assessment.	22
3.5 The Soil and Water Assessment Tool Modelling	23
3.6 Data Preparation for SWAT Model	24
3.6.1 Land Use and Land Cover	24
3.6.2 Land Topography Data	25
3.6.3 Soil Data.....	25
3.6.4 Climate Data	26
3.6.5 Soil and Water Assessment Tool Procedure	26
3.5.6 SWAT Analyses.....	27
CHAPTER 4: RESULTS AND DISCUSSION	34
4.1 SWAT Model Map Results.....	34
4.2 SWAT Stream Flow.....	36
4.3 LULC Change Detection Maps.....	37
CHAPTER 5: CONCLUSIONS AND RECOMMENDATIONS	41
5.1 Introduction.....	41
5.2 Conclusions.....	41
5.3 Recommendations.....	41
REFERENCES.....	43

LIST OF FIGURES

Figure 1: The integration of Aquifers and Surface Water.	7
Figure 2: Rainfall Distribution Annually.....	16
Figure 3: Study Area Hydro Geology	17
Figure 4: The Study Area Map.	18
Figure 5: Methodology Flowchart	20
Figure 6: Accuracy Assessment for LULC 2013.....	22
Figure 7: Accuracy Assessment for LULC 2017.....	22
Figure 8: Accuracy Assessment for LULC 2021.....	23
Figure 9: ArcSWAT Set Up.....	27
Figure 10: Watershed Delineation.	27
Figure 11: Land use Definition in SWAT.....	28
Figure 12: Soil Data Definition in SWAT.	29
Figure 13: Slope Definition in SWAT.	30
Figure 14: Hydrological Response Units Thresholds.	31
Figure 15: Weather Data Definition in SWAT.	31
Figure 16: Write SWAT Database Tables.	32
Figure 17: Set Up and Run SWAT Model Simulation.	33
Figure 18: Successful SWAT Run check.....	33
Figure 19: SWAT Model Maps, DEM and LULC	35
Figure 20: SWAT Model Maps, Soils and Slope Distribution	35
Figure 21: SWAT Model Maps, HRUs and Final Map	36
Figure 22: SWAT Model Streams Flow INcms.	37
Figure 23: SWAT Model Streams Flow OUTcms	37
Figure 24: LULC Trends	38
Figure 25: LULC Graph for 2013.....	38
Figure 26: LULC graph for 2017.....	39
Figure 27: LULC graph for 2021.....	39
Figure 28: Overall LULC Graph.....	40

LIST OF TABLES

Table 1: Study Area Hydro Geology	17
Table 2: Methodology Datasets	18
Table 3: Classification Scheme for the classification and Change Detection Procedure.	21
Table 4: Data Types Used in SWAT and their Sources.....	24
Table 5: Land Use Codes and Percentage of Coverage.	24
Table 6: Soil Textures and Percentage of Distribution.	25
Table 7: Soil Data Codes.	25
Table 8: SWAT Run check results.....	34
Table 9: Positive and Negative Impacts.....	40

ABBREVIATIONS AND ACRONYMS

ASAL	Arid and Semi-Arid Lands
CHIRPS	Climate Hazards Group InfraRed Precipitation
DEM	Digital Elevation Model
FAO	Food and Agriculture Organisation
GDP	Gross Domestic Product
GIS	Geospatial Information Systems
HRU	Hydrological Response Units
LULC	Land Use and Land Cover
OLI	Operational Land Manager
SWAT	Soil and Water Assessment Tool
TAWSB	Tanathi Water Services Board
USGS	United States Geological Survey

CHAPTER 1: INTRODUCTION

1.1 Background Of the study

The implementation of GIS based models and remote sensing provides new opportunities for solving the present and futuristic geo-related problems. GIS based models are crucial to cope with the new geo-related problems like; floods, droughts, land use planning and management and effects of global warming, especially the change of rainfall patterns and temperature. Those influences are very complex and therefore GIS-based models represent as efficient tools to overcome difficulties in understanding and management of river basins (Hoffmann et al., 2009).

Water and soil are indispensable resources needed in all aspects of human activities. Availability of water both positively and negatively affects human and animal habitats, economic prosperity, food supply, human migration, and domestic and international relations. The rapid population growth in both rural and urban regions coupled with climate change, puts a lot of pressure on water resources, especially in water scarce regions (Silva et al., 2022). The geographical locations and timing of water users influence the spatial and temporal distribution of water availability across river basins.

Many arid and semi-arid regions face limited availability of water resources which affects its potential demand. In this era of climate change, it is essential to understand dynamics of water availability and how to manage it sustainably (Ires, 2021). It is important to also develop methods and quantify water for agricultural, domestic, and industrial sectors.

One of the best methods to quantify the water availability and soil types is the use of GIS based modelling with Soil and water assessment tools and reliable datasets (Omani et al., 2007). The integrated approach of using SWAT in GIS modelling and remote sensing proves to be fruitful in determining the best practices and policies to be adhered in a river basin management.

1.2 Problem Statement.

The Thwake River basin has experienced a marked decline in rainfall over the years (Langat et al., 2017). The inhabitants of the region trek long distances to fetch water for both domestic and farm use despite having seasonal rivers in the basin where reservoirs can be constructed to store water during the rainy season for use in the dry seasons.

Due to increase in population growth in the region, the inhabitants have cleared most of the land cover to find space for land use; settlement and farming, hence the basin is rapidly changing from semi-arid to arid. Small scale farmers in the area face loss of crops due to declining annual rainfall and harsh climatic conditions experienced in the region.

Improper irrigation methods and growing of crops along the basin valleys has led to pollution in the lower parts of the basin. Wastewater and sewage from the major towns and market centres in the region is drained in to the rivers, which flow in the basin making it a hazard for water borne diseases and water stagnation. Most of the water dams and reservoirs are filled with flood water silt over time during the rainy seasons causing less water collection for the next rain season.

Poorly constructed road networks in most rural parts of the basin destruct the traders and business people who transport food crops such as vegetables, cereals and fruits to the major towns and rural market centres, especially during the rainy seasons. Poorly constructed bridges across the basin rivers are destroyed by heavy rains that cause massive water flood water flow in the basin river valleys inhibiting transportation of goods from one region to another, affecting the economic development of the inhabitants.

The application GIS based modelling and Remote sensing is the most standard technique which can be applied for the river basin management and determine the best framework policies to be implemented for water management (Jasodani, 2020).

1.3 Justification for the study

There have been various efforts led by researchers in the fields of hydrology, environmental management, geography, geology, and geospatial engineering to determine the best management alternatives for river basins. The emphasis of these efforts has mostly been focused within river basins in highlands. However, one of the challenges has always been proper understanding of the areas of study and suitable methodology frameworks and availability of datasets for the success of these studies.

GIS based models integrated with Remote sensing methodologies in river basin hydrological modelling present an opportunity for proper and sustainable managements tasks for water resources and land use. GIS techniques integrate crucial frameworks from conceptualization, analysing, assessment, planning, visualization, and output of suitable river basin hydrological results, especially for water resources.

1.4 Objectives.

1.4.1 Overall Objective.

To evaluate the impact of Land Use and Land Cover changes on the hydrological processes in the Thwake River Basin using GIS-Based models and remote sensing techniques.

1.4.2 Specific Objectives.

In specific the objectives included the following:

1. To simulate the hydrological processes in the Thwake River Basin using the Soil and Water Assessment Tool (SWAT) model.
2. To perform a change detection analysis of LULC in the Thwake River Basin using GIS and remote sensing data.
3. To assess the impact of LULC changes on the hydrological processes in the Thwake River Basin.

1.5 Research Questions.

1. What is the extend of the River Basin?
2. What is the trend of the land use and land cover?
3. What are the positive and negative impacts of land use and land cover on the basin?

1.6 Scope of work.

The project involved the hydrological modelling of the Thwake River Basin using the Soil and Water Assessment tool. The model generated the basin area, watersheds, pour points and the longest stream reach. Sentinel satellite imagery obtained from USGS website for the year 2022 was classified and used and input in the SWAT land use category, Digital Soil map clipped to the area of study was used in the soil category and the Source DEM which was used in the generation of the hydro networks. Change detection trends was carried out for the years 2013, 2017 and 2021 using Landsat 8 imagery. Classification accuracy was done for the land use and land cover classified images.

The project focused on Thwake Basin which covers, which covers Machakos County and Makueni County. The study region which is a river basin was selected due to water shortage issues, food insecurity, unfavourable changing weather patterns and climatic conditions experienced in the river basin over the past recent years.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

A river basin area should be seen as interdependent and interconnected, with elements and components as well as various activities within a river basin. These activities can affect water levels or can have an impact on other areas such as environmental, social and economic conditions (Molle et al., 2007).

Therefore, it is necessary to choose a comprehensive, systematic, and coordinated process of conceptualization, planning, control, organization, assessment and management within the river basin, which will be based on the knowledge that the water in the landscape-topographical and ecological perspective is one of the primary components of the landscape and also structure of topography. Its use depends on the way of use and the policies put in place. Such an approach reflecting the multidisciplinary nature in the context of the overall social, environmental and economic development and other interests relating to the use and protection of water resources in river basins such as; in the field of water supply and sewerage systems, agriculture and irrigation, industrial use, residential development, water works, recreation, fishing and other activities, while emphasizing the need for coordination between sectors and proposing the need for adoption of different systems planning and management can be said to be an integrated river basin management (Ismail & Section, 1997).

The effectiveness of an integrated river basin management with the use of GIS based models and Remote Sensing occur during extreme hydrological events such as river basin flood waters and droughts (Principe & Blanco, 2012). Flooding occurs when a river's discharge exceeds its channel's volume causing the river to overflow onto the areas surrounding the channel called a flood plain. The increase in discharge can be triggered by several events such as prolonged rainfall. When it rains for a long time, the ground becomes saturated and the soil will no longer be able to store water leading to increased surface run off. In semi-arid and arid areas, floods are mostly caused by brief periods of heavy rainfall. The rain water won't be able to infiltrate fast enough and therefore the water will instead enter the river via surface run off which leads to sudden and large increase in the rivers discharge which then results to flash floods (Jackson, 2014).

Most of the researches in the 1980s uncovered more the 150 definitions of droughts; the definitions also reflect differences in regions, needs, disciplinary approaches and impacts on the society.

Wilhite and Glantz categorized the definitions in terms of four basic approaches to measuring drought: hydrological drought, meteorological drought, agricultural drought and socio-economic drought. The first three deal with ways to measure drought as physical phenomenon while the last deals with drought in terms of supply and demand, tracking the effects of water decline as it ripples through social and economic systems (Wilhite & Glantz, 2019).

2.2 Overview of GIS based modelling and Remote Sensing application in river basin management.

River basin management reduces excess streamflow, thereby reducing soil erosion and non-point source pollutants in the form of nutrients from agriculture dominated areas. Human activities like burning fossil fuels and land use changes help increase greenhouse gas concentrations that lead to climate change. Climate change has shown its impacts on water resource availability and management throughout the world (Pandey et al., 2021). Those changes affect the hydro logical cycles critical components such as streamflow. As cited by Pandey, the impact of climate and land use and land cover change can be modelled in the Soil and Water assessment Tool. GIS based modelling simplifies the process and evaluates it for proper decision and policy making.

(Jasodani, 2020) cites that GIS based modelling using Soil and Water Assessment Tools is essential in checking and enhancing a watershed development with positive externalities. Water, worldwide is fast becoming a scarce resource due to increase in world population, industrialization, urbanization and uncertain rainfall. For instance, India though having immense water resources is lately facing water stress in several parts of the country. Jasodani undertook research on Watrak river basin, a part of Sabarmati River basin in Gujarat, the western state of India. SWAT Hydrological model building required certain spatial and non-spatial datasets which included DEM, Land Use raster and soil map clipped to the area of study. The statistical parameters of the model were found to be in closer range showing that the model is still minimally acceptable and can reasonably predict the effects of varying land management practices and conservation measures on the water balance within the basin.

2.3 Types of Basins.

2.3.1 River Drainage Basin

There is a general consensus among researchers including, who have defined a river basin as an area of land where precipitation collects and drains off into a particular point along a channel

network or depression (Goetz et al., 1978) . The basin is the basic unit for a hydrological study because the input and output can be quantified and accessed; the basic input is precipitation and largely rainfall in the humid region while the output or the response is the runoff. Further, these researchers have argued that rainfall-runoff relationship provides insights into a basin's input-output relationship, and consequently, a basin's behaviour and an indicator of the basin's status of health.

A river basin is an open system and therefore it receives inputs of wastes, seepages and also debris from the anthropogenic and natural activities that are within the confinement of the basin and are capable of influencing the quality of the water. Land cover, topography, geology, shape and size of a river basin are capable of posing a significant influence on the river basin response to precipitation input.

2.3.2 Structural Basins

A structural basin is a large-scale structural formation of rock strata formed by tectonic warping of previously flat lying strata. Structural basins are geological depressions, and are the inverse of domes. Some elongated structural basins are also referred to as synclines. Structural basins may also be sedimentary basins, which are aggregations of sediment that filled up a depression or accumulated in an area; however, many structural basins were formed by tectonic events long after the sedimentary layers were deposited (DBpedia, 2022).

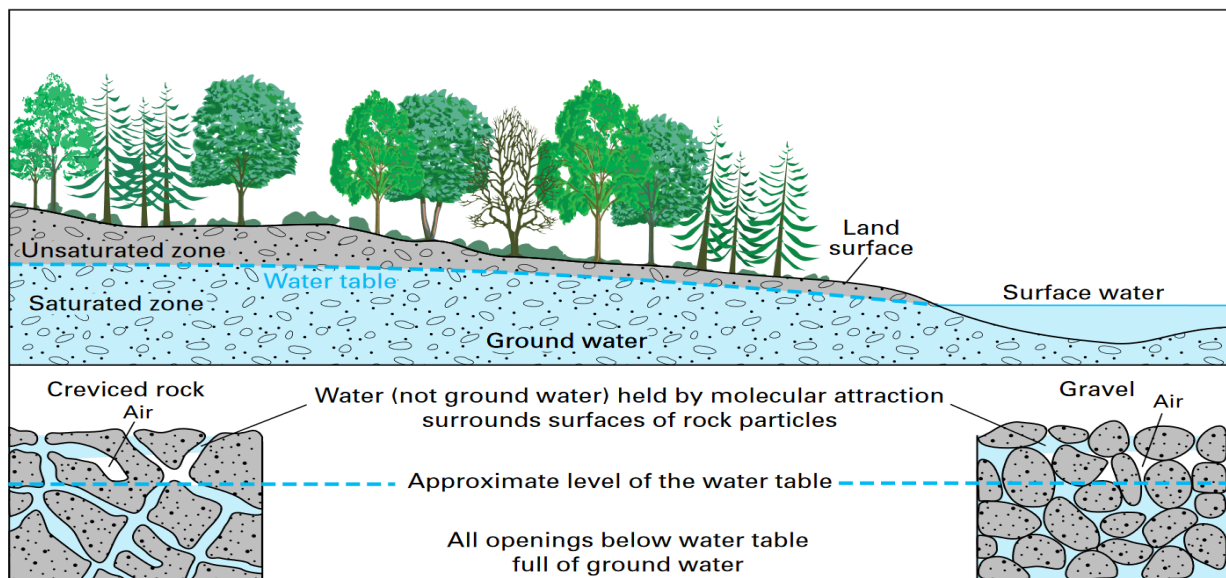
Some structural basins are known as endorheic basins. Endorheic basins have internal drainage systems. This means they don't have enough water to drain to stream, lake or ocean. The water that trickles in to these types of basins evaporates or seeps into the ground (Goetz et al., 1978)

2.3.3 Ocean Basins

In hydrology, an oceanic basin (or ocean basin) is anywhere on earth that is covered by seawater. Geologically, ocean basins are large geologic basins that are below sea level. There are five major ocean basins, coordinating with the major oceans of the world: the Pacific basin, the Atlantic basin, the Arctic basin, the Indian basin and the Southern basin. Many smaller basins are often considered oceanic basins, such as the North Aleutian basin, between the Pacific and Arctic Oceans. Tectonic activity constantly changes ocean basins. Seafloor spreading and subduction are most important types of tectonic activity that shape ocean basins (Goetz et al., 1978).

2.4 The Integration of Aquifers and Surface Water

Surface water seeps into the ground and recharges the underlying aquifer – groundwater discharges to the surface and supplies the stream with baseflow. Groundwater and surface water physically overlap at the groundwater and surface water interface through the exchange of water and chemicals (Ezugwu & Apeh, 2017). This exchange is a critical part of the hydrologic cycle. Surface water supplies recharge to the underlying aquifer, where the groundwater can remain in storage for days, months, years, centuries or even millennia. Eventually the ground water discharges back into the stream. Depending on how much time the water spends underground, and the geochemical conditions within the aquifer, the quality of the original recharge water can also undergo profound changes before it discharges at the surface (Water Resources, 2019).



How ground water occurs in rocks.

Figure 1: The integration of Aquifers and Surface Water. Source (<https://d9-wret.s3.us-west-2.amazonaws.com/assets/palladium/production/s3fs-public/thumbnails/image/wss-gw-how-groundwater-occurs-diagram.png>)

An aquifer is a body of porous rock or sediment saturated with groundwater. Groundwater enters an aquifer as precipitation seeps through the soil. It can move through the aquifer and resurface through springs and wells. There are two types of aquifers:

1. Confined aquifers.
2. Unconfined aquifers.

A confined aquifer is an aquifer below the land surface that is saturated with water. Layers of impermeable material are both above and below the aquifer, causing it to be under pressure so that when the aquifer is penetrated by a well, the water will rise above the top of the aquifer.

An unconfined aquifer also referred to as a water table is an aquifer whose upper water surface is at atmospheric pressure, and thus is able to rise and fall. Water table aquifers are usually closer to the earth's surface than confined aquifers are, and such are impacted by drought conditions sooner than confined aquifers.

Many different types of sediments and rocks can form aquifers, including gravel, sandstone, conglomerates and fractured limestone. Aquifers are sometimes categorized according to the type of sediments or rocks of which they are composed. Much of the water used for domestic, agricultural or industrial purposes is groundwater. Most groundwater, including a significant amount of our drinking water, comes from aquifers. In order to access this water, a well must be created by drilling a hole that reaches the aquifer. While wells are manmade points of discharge for aquifers, they also discharge naturally at springs and in wetlands.

Groundwater can become depleted if we use it at a faster rate than it can replenish itself. This is brought about by the increasing population growth and advanced agricultural methods for irrigation. Groundwater can become contaminated when an excessive number of pesticides and herbicides are sprayed on agricultural fields, septic tank leaks, or landfills are improperly lined or managed and toxic materials seep through the soil into the aquifer.

2.5 Water table

The water table describes the boundary between water saturated ground and unsaturated ground. Below the water table, rocks and soil are full of water. An area's water table can fluctuate as water seeps downward from the surface. It filters through soil, sediment and rocks. This water includes precipitation, such as rain and snow. Irrigation from crops in a water basin and other plants may also contribute to a rising water table. This seeping process is called saturation. Sediment or rocks that are full of water are saturated. The water table sits on the zone of saturation, or the phreatic zone. The area above the water table is the vadose zone. Water tables often follow the topography of a river drainage basin, or upward and downward tilts, of the land above them (National Geographic, 2022).

In addition to topography, water tables are influenced by many factors, including geology, weather, ground cover and land use. Geology is often responsible for how much water filters

below the zone of saturation, making the water table easy to measure. Water tables are also influenced by weather. They will be usually be higher in rainy seasons. Ground cover can contribute to an areas water table. The spongy, absorbent vegetation in swamps is saturated at least part of every year. Water tables in swamps are nearly level or even higher than the surface. Land use can also influence an areas water table. Urban areas mostly have impervious surfaces, impervious surfaces prevent water from seeping in to the ground below. Instead of entering the areas zone of saturation, water becomes runoff.

2.6 Climate and Climate Change in Kenya

Climate change in Kenya is increasingly impacting the lives of citizens and the environment. Climate change has led to more frequent extreme weather events like droughts which last longer than usual, irregular and unpredictable rainfall, flooding and increasing temperatures. The effects of these climatic changes have made already existing challenges with water security, health, food security and economic growth even more difficult (Alcayna et al., 2021). Harvests and agricultural production which account for about thirty-three percentage of total Gross Domestic Product (GDP) are also at risk.

The increased temperatures, rainfall variability in arid and semi-arid areas, and strong winds associated with tropical cyclones have combined to create favourable conditions for the breeding and migration of pests. An increase in temperature of up to 2.5 °C by 2050 is predicted to increase the frequency of extreme events such as floods and droughts.

Hot and dry conditions mostly in Arid and Semi-Arid Lands (ASALs) make droughts or flooding brought on by extreme weather changes even more dangerous. These factors impact at risk populations like marginalized communities.

2.7 Impacts of Climate Change

2.7.1 Impact on the Natural Environment

The temperature anomalies were positive every year since 2000 with respect to the climatological mean of the years 1981 to 2010 according to satellite data. Analysis of climate trends in Kenya's Arid and Semi-Arid (ASAL) areas shows an increase in temperature and a decrease in rainfall between 1977 and 2014. Climate change impacts are predicted to be particularly pronounced in ASALs where the economy and rural livelihoods are highly dependent on climate-sensitive activities, such as rainfed cultivation. Increased temperatures, rainfall variability and strong winds associated with tropical cyclones have combined to create

favourable conditions for the breeding of insects and pests. For instance, in early 2020 some parts of Kenya and neighbouring East African countries faced massive swarms of locusts.

2.7.2 Impact on Forested Watersheds

Forested watersheds improve water quality and enhance water storages, naturally regulate streamflow. Reduce flood damages and stormwater runoff, replenish groundwater and provide a myriad of other benefits. Significant changes in temperature and precipitation present challenges to ensuring long term ecosystem health, particularly in sensitive and unique riparian areas, and may amplify existing challenges related to infrastructure, stormwater quality and quantity (Marshall & Randhir, 2008).

Climate change will not affect all parts of a watershed the same way. The local response and vulnerability of a watershed to regional climate changes will be influenced by soils, topography, forest management and current and past land uses. Warmer temperatures and variable precipitation will have direct impacts on the water budget and the hydrology of forested watersheds; affecting the timing, distribution and seasonal availability of precipitation. Forest ecosystem vulnerability assessments can help the managers characterize site vulnerability in forested watersheds.

2.7.3 Impact on River Drainage Basins

Major rivers worldwide have experienced dramatic changes in flow, reducing their natural ability to adjust to and absorb disturbances. Given the changes in global climate and water needs, this may create serious problems, including loss of native biodiversity and risks to ecosystem, and humans from increased storm water flooding or water shortages (Palmer et al., 2008). Healthy, free flowing rivers respond to changes in land use and climate change through dynamic movements and flow adjustments that buffer against climatic changes. However, river drainage basins are sufficiently impacted that their ability to absorb disturbances, such as changes in discharge and water stress, is severely limited. A global analysis of the potential effect of climate change on river drainage basins indicates that rivers impacted by dams or extensive developments will require more management interventions to protect ecosystems and people than basins with free-flowing rivers. Specific, proactive restoration, rehabilitation and management actions are recommended to enhance the resilience of riverine ecosystems and minimize impacts (Palmer et al., 2008).

2.8 River Basin Management Projects in Thwake River Basin

2.8.1 Thwake Dam Water Project

Thwake Multipurpose Dam is a project under the Tanathi Water Services Board (TAWSB) and entails harnessing of the fluctuating flows of the Athi River and the seasonal water from Thwake River. The dam will provide water supply for domestic, livestock, irrigation, hydropower and even industrial activities in the beneficiary districts. It will largely serve areas in Makueni and neighbouring districts but also certain parts of Kitui based on topography (Rofifah, 2020).

Thwake Dam was first proposed for construction in 1953 during the colonial days and again in the 1980s. However, the project was never undertaken. Significant numbers of intervention projects in water and sanitation have been undertaken in Makueni County and other area in the southern part of Eastern province, but the problem of water shortage continues to worsen particularly with regard to its sustainable availability. The proposed multi-purpose dam is been designed to serve the greater part of Makueni County with Kalawa, Kathonzweni, Kibwezi and Makindu as the main target areas. Other areas to benefit include sections of Wote, Nguu and Mutio Andei areas (Multi-purpose & Report, 2013).

Residents in Makueni County are forced to walk distances of up to 10km in search of water for livestock and domestic use. The little water found is mostly of inferior quality effectively risking the people's health and hygiene. This is clearly depicted in the reported cases of water borne ailments; diarrhoea, intestinal worms, bilharzias and skin problems. Therefore, the proposed dam project is a necessary intervention that will address water scarcity and in turn open up area for development and improve the social and economic status of the involved areas. The dam will open up the area for development in terms of activities such as fishing, tourism and the fact that the dam axis may provide a causeway that will open up the area by providing social and economic access between Makueni, Machakos and Kitui Counties. The overall ecosystem of the project area stands to get transformed to the benefit of the communities.

2.9 The Application of GIS-Based Models and RS in River Basin Management

2.9.1 Using GIS-Based Models to Manage River Basins.

(High Point NC, 2019) argue that a geographic Information System (GIS) is a computer-based tool for mapping and analysing things that exist and events that happen on earth. Further, GIS technology integrates common databases operations such as query and statistical analysis with

the unique visualization and geographic analysis benefits offered by maps. Moreover, the researchers explain that, these abilities distinguish GIS from other information systems making it valuable to a wide range of usage; especially in water resource management in river basins for explaining events such as droughts, predicting outcomes of floods and planning strategies.

Geographic Information Systems simulate a given geographic space in a computational environment allowing for storage, mapping and analysing large amounts of data. GIS were converted into a powerful tool regional natural resource assessment, as it permits a speedy integration and representation of several biophysical attributes (Olaf Bastian, 2000) from various origins such as topography, hydrology, geology and remote sensing.

Many authors argue that for a successful river basin management, remote sensing data is crucial to the study. Remote sensing is concerned with acquiring information about the earth's land and water surfaces with reflected or emitted electromagnetic energy. Sensors fixed to a platform detect and record electromagnetic energy from target areas in the field of view of the sensors' instrument. When a satellite imagery is classified, it can be assigned classes for land cover and land use for a study area; agricultural, transport, vegetation, riverine, build up areas and also farmlands. Remote sensing imagery and geographical information are useful for land use land cover classification, land degradation and soil erosion.

2.10 The Soil and Water Assessment Tool

Soil and Water Assessment Tool (SWAT) developed by USDA is a physically based distributed model which helps in predicting and simulating the land phase of hydrologic cycle as well as the impact of land management practices on water, sediments in watersheds in daily time step, monthly and yearly. SWAT model is designed to simulate the water and sediment movement from individual watersheds through the river basin systems (Omani et al., 2007). SWAT model is flexible enough to include reservoirs, tanks, check dams off stream as well as model on stream in simulating the land phase. SWAT simulates eight important components which include: hydrology, erosion, sediment transport, weather, nutrients, crop growth, pesticides, agricultural management and soil temperature. Major hydrological processes simulated by the model are infiltration, surface runoff, channel routing, shallow aquifer, deep aquifer, percolation and evapotranspiration. The major advantage of the Soil and Water Assessment Tool is that, unlike conventional simulation models, it needs little calibration to run and hence can be used on ungauged watersheds.

The SWAT model outputs incorporate water balance components which may include: surface run off, recharge, percolation, lateral flow, evapotranspiration and sediment yield, at the scale of each watershed and at daily, monthly or annual time steps. Each watershed is first divided into sub basin and then into hydrologic response units (HRUs) based on the land use, soil distribution and slope. The water storage components are soil profile, shallow aquifer, deep aquifer and snow cover. A daily water budget is established for each HRU based precipitation, surface runoff, evapotranspiration, base flow, percolation and soil moisture change. Each HRU for each sub basin and watershed computes runoff using Soil Conservation Services developed Curve Number (CN) procedure. Spatial datasets used in the SWAT model include DEM, FAO soils, land use and land cover.

2.11 Critique of Previous Research Work

The literature reviewed has highlighted the use of GIS based modelling and Remote Sensing in all river management, assessment and mapping. Specific instances of how GIS modelling and remote sensing has been used in river basin management have been given, for instance the application of SWAT model for upper Sabarmati basin in India (Jasodani, 2020) . However, there was no complete document or presentation that assisted in comprehensive analysing of factors necessary in a river basin management using both GIS models and Land Use and Land Cover change detection.

2.12 Research Gaps

A number of research works on the use of GIS based models and remote sensing in river basin management have been done in various countries. However, there was no related research work carried out in Thwake River Basin with regard to river basin management using GIS Based models and remote sensing. While calibration of GIS based models is a tedious work, more work still needs to be done by way of calibrating various parameters so as to get higher quantitative parameters measuring the goodness of fit. This way the predictions made through the model would be further be more conclusive. In addition, more research needs to be conducted using more accurate data, which yield much better and reliable results, especially in management of river basins in ASAL regions. This will go a long way in ensuring water sustainability, food security, social and economic growth and ecosystem management.

CHAPTER 3: MATERIALS AND METHODS

3.1 Introduction

This chapter entails the systematic and conclusive description of the stages involved in the processing of the raw datasets to provide insightful and reliable results, thus meeting the objectives set for the research. The chapter also includes the methodology framework, SWAT modelling techniques and remote sensing techniques used to derive meaningful results from the raw data which include DEM, Satellite imagery and Digital Soil map.

3.2 The Study Area

Thwake River Basin is situated in between Machakos and Makueni counties in the Eastern region of Kenya. The basin covers an area of 1637.65 km² and is geographically situated between longitudes 37°10'0" East and 37°30'0" and latitudes 1°20'0" South and 1°40'0" South. The main rivers in the basin include Thwake River, Mwanja River, Kalusi River, Maruba River, Ikiwe River and Kaiti River. Maruba dam is also located in the basin. The basin covers almost 30 wards in Makueni and Machakos Constituencies. The major river is Thwake River which drains to Thwake dam and a tributary to Tana River. The basin is made up of 6,310 sub basins.

The total population of the basin was 67,007 according to the 2009 census. The population growth has continuously risen over the years according to the 2019 census. Majority of the inhabitants of the basin are the Akamba community. Most of the inhabitants are small scale farmers and traders. Other communities present in the region are engage in either self-employed enterprises or employment in the two counties.

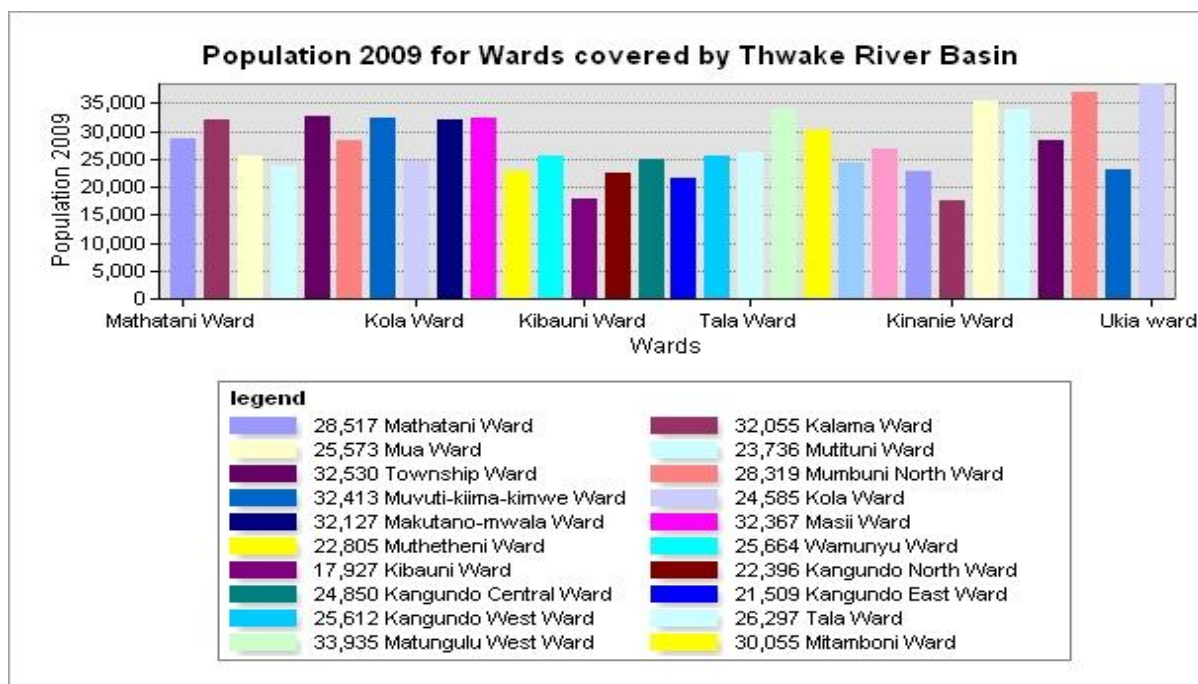


Figure 2: 2009 Population Graph.

The basin experiences Tropical rainfall due to its location with the tropical and semi-arid zones of Kenya. The rainfall ranges from 1,300mm – 500mm annually. In terms of altitude, the basin has a highest altitude of 2,152m and lowest altitude is 561m above sea level. Agriculture is the main economic activity in the area. Some of the major crops grown include coffee, maize, beans, peas and kales. Fruits are also part of the major source of income. Some of the major grown fruits include paw paws, mangoes, lemons and oranges.

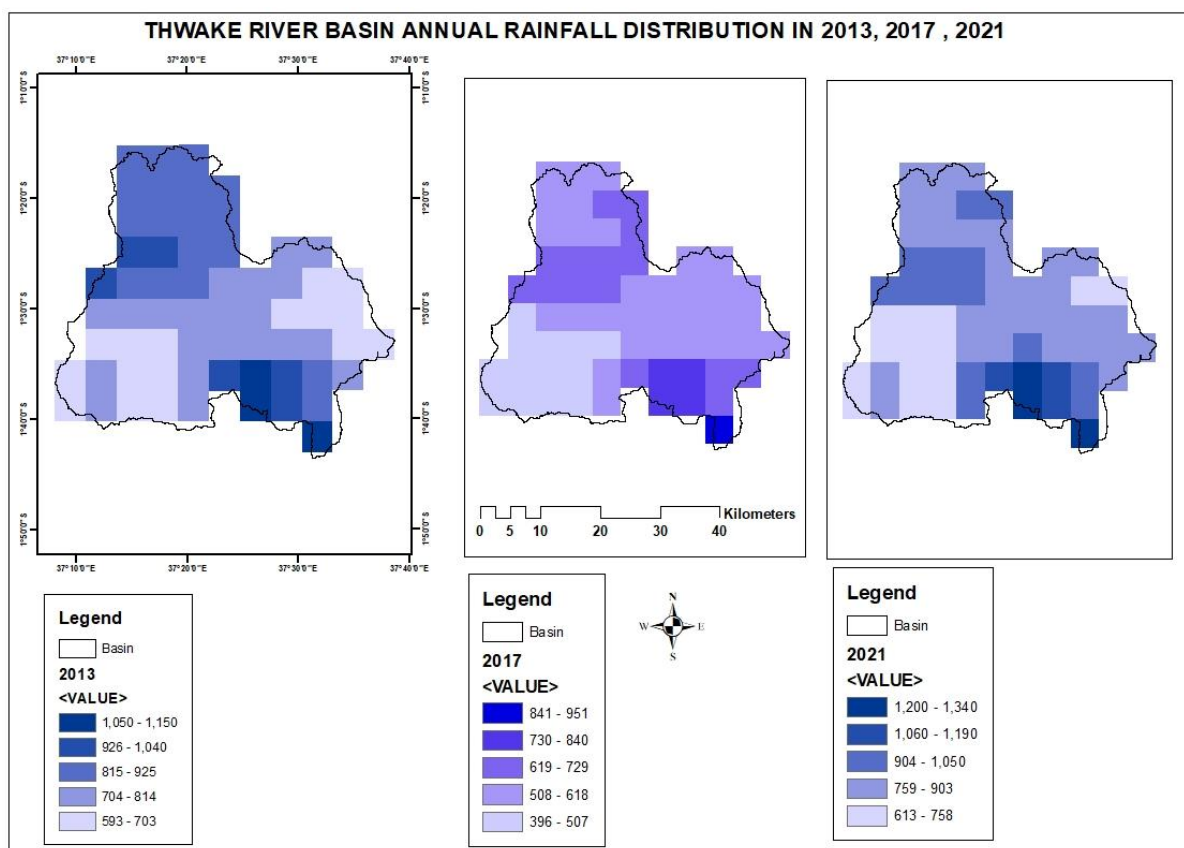


Figure 2: Annual rainfall distribution.

The basin a wide road network and most parts of the basin are easily accessible. Some of the major highways include the Machakos Kitui highway, Machakos Wote Highway and Machakos Kangundo Highway. The major market centres are Machakos Town, Masii, Kangundo, Mbooni, Kiteta, Miu, Kaani, Katelembo, Makutano and Mutituni.

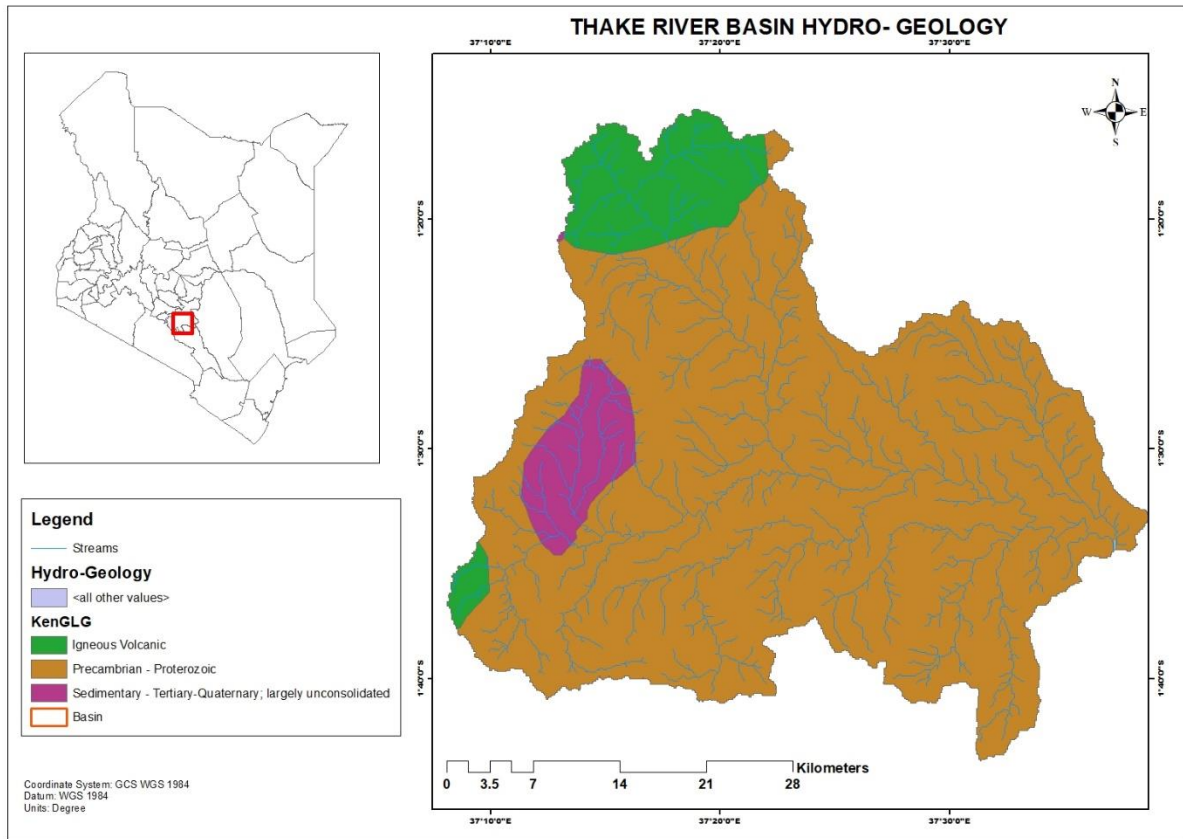


Figure 3: Study Area Hydro Geology

The Geology, hydro geology, aquifer types and their productivities are summarised in the table below.

Table 1: Study Area Hydro Geology

Geology	Hydro Geology	Aquifer Type and Productivity
Igneous Volcanic	Igneous Volcanic	Volcanic Moderate
Precambrian - Proterozoic	Precambrian - Archaean	Unconsolidated/Semi consolidated Intergranular – Moderate to high
Sedimentary – Tertiary – Quaternary; Largely unconsolidated	Sedimentary – Tertiary – Quaternary; Largely unconsolidated	Basement – Low to Moderate

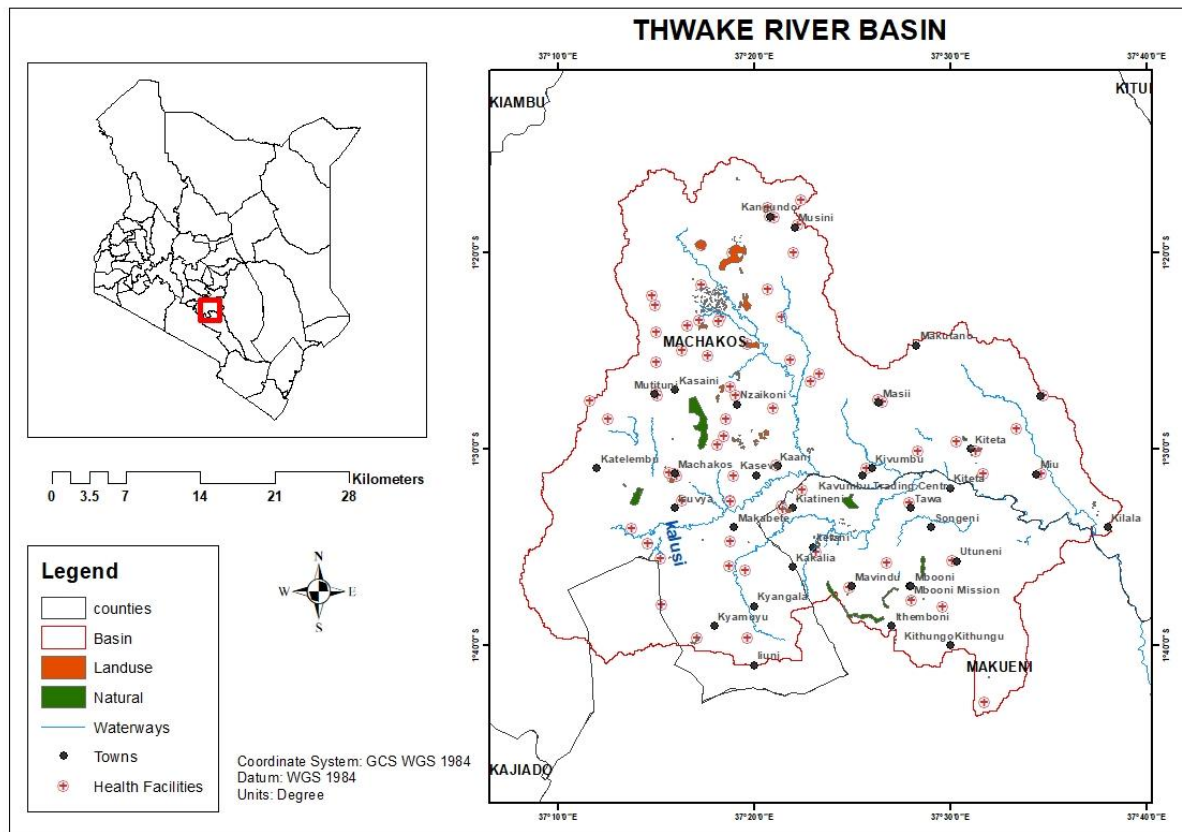


Figure 4: The Study Area Map.

3.3 Methodology Datasets Acquisition.

The datasets for this research were the Digital Elevation Model (DEM) with a resolution of 30m, Sentinel satellite imagery for the year 2022 with a resolution of 20m, Landsat 8 OLI imagery for the years 2013, 2017 and 2022 with a resolution of 30m, FAO digital soil map, hydrogeology data which had aquifers and hydrology and climate data for precipitation. The DEM and Landsat 8 imagery were downloaded from USGS website, the digital soil map was acquired from the FAO website, climate data was acquired from CHIRPS website and hydrogeology data was acquired from Africa Groundwater Atlas. A summary of the datasets, their characteristics where applicable is as shown in the table below.

Table 2: Methodology Datasets

Dataset	Dataset Characteristics	Source
DEM data	Resolution 30m	USGS
Sentinel 2A,2B	Year 2022, Bands 11, Resolution 20m	USGS

Landsat 8 OLI	Years 2013, 2017, 2021, USGS Bands 8, Resolution 30m	
FAO soil data	Soil types, Digital Soil Map	FAO website
Hydrogeology data	Hydrology and aquifers	Africa Groundwater atlas
Climate Data	Precipitation	CHIRPS

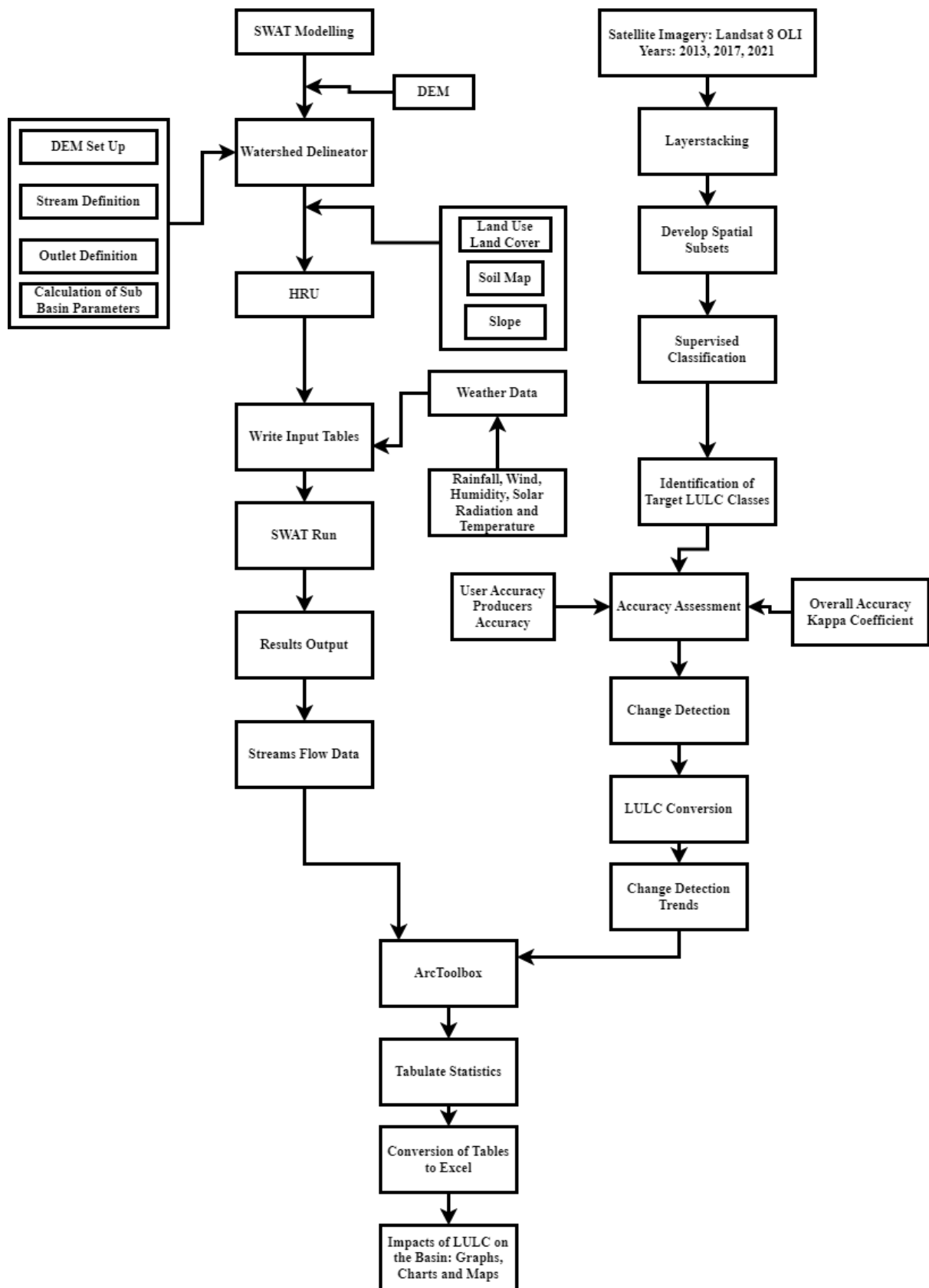


Figure 5: Methodology Flowchart

3.4 Data Analysis

3.4.1 Data Acquisition and Preparation.

Landsat imagery was used to determine LULC change, while dates were selected based on data quality, data availability, and the dry season. Three Landsat 8 OLI imageries were acquired for the years 2013, 2017 and 2021 from the USGS for the Thwake River Basin Area of study. These data were used for the generation of LULC maps fed into on a GIS. The software packages, ERDAS imagine 2011, ENVI and ArcGIS 10.5 were employed at various stages of analysis. All the images have the same spatial resolution of 30m, but data were sensed by different sensors at different times of the year. After mosaicking the Landsat scenes by date, the resulting three images were clipped to the study area.

3.4.2 Classification and Change Detection

A supervised classification technique was adopted to digitally categorize each Landsat image, since this technique has been showing to perform better in the case of spectral variability of individual cover types. For this study, a priori identified six (5) land use and land cover classes were subsequently used in the classification scheme. These were Vegetation, Build Up, Bare land, Agriculture and Water.

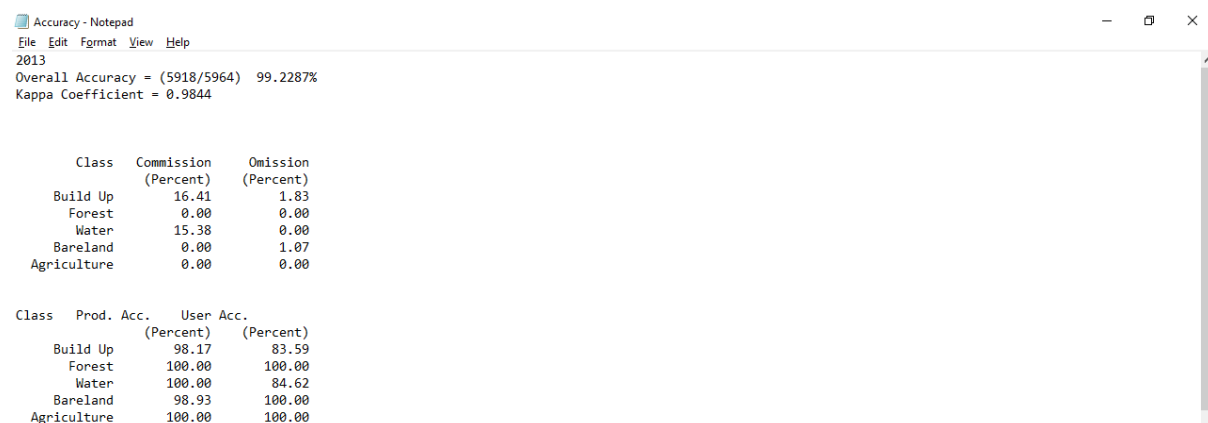
Table 3: Classification Scheme for the classification and Change Detection Procedure.

Indicator	Class Name	Description
Vegetation	Forest, Evergreen Forests	Areas under forest mainly on the higher altitudes.
Build-Up Area	Urban area, Settlements	Area's characteristic of highly developed towns, urban areas and other built-up areas.
Bare land	Bush, Savannah, grassland	Areas vegetated with grass and with relatively low occurrence of shrubs.
Agriculture	Farmlands	Areas under subsistence farming or areas under commercial large-scale farming

Water	Hydrology, Riverine	Dam reservoir, rivers and streams.
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3.4.3 Accuracy Assessment.

Accuracy assessment is essential for individual classification if the classification data is to be useful in change detection. Normally, classification accuracy is carried out by comparison of two datasets: one based on reference information, which is referred to as ground truth, and the other being the result of the analysis of remotely sensed data. Overall, LULC classification levels for the three imageries ranged from 97% to 99%, with kappa indices of agreement ranging from 0.94 to 0.98. Accuracies per individual LULC class, that is the user accuracy and the producer accuracy are shown in the figures below. These values of Kappa for the three classification results are satisfactory for the study area because they satisfy the minimum 85% accuracy, as stipulated by Anderson's classification scheme.

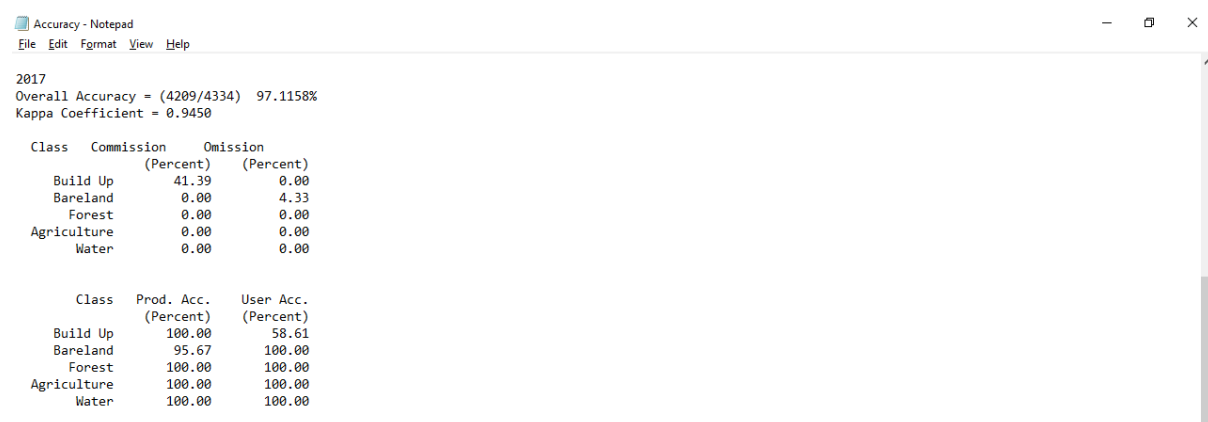


2013
Overall Accuracy = (5918/5964) 99.2287%
Kappa Coefficient = 0.9844

Class	Commission (Percent)	Omission (Percent)
Build Up	16.41	1.83
Forest	0.00	0.00
Water	15.38	0.00
Bareland	0.00	1.07
Agriculture	0.00	0.00

Class	Prod. Acc. (Percent)	User Acc. (Percent)
Build Up	98.17	83.59
Forest	100.00	100.00
Water	100.00	84.62
Bareland	98.93	100.00
Agriculture	100.00	100.00

Figure 6: Accuracy Assessment for LULC 2013.



2017
Overall Accuracy = (4209/4334) 97.1158%
Kappa Coefficient = 0.9450

Class	Commission (Percent)	Omission (Percent)
Build Up	41.39	0.00
Bareland	0.00	4.33
Forest	0.00	0.00
Agriculture	0.00	0.00
Water	0.00	0.00

Class	Prod. Acc. (Percent)	User Acc. (Percent)
Build Up	100.00	58.61
Bareland	95.67	100.00
Forest	100.00	100.00
Agriculture	100.00	100.00
Water	100.00	100.00

Figure 7: Accuracy Assessment for LULC 2017.

2021

Overall Accuracy = (3032/3128) 96.9309%
Kappa Coefficient = 0.9522

Class	Commission (Percent)	Omission (Percent)
Build Up	4.80	22.05
Forest	2.36	0.00
Water	0.00	0.00
Bareland	4.25	0.78
Agriculture	0.00	1.61

Class	Prod. Acc. (Percent)	User Acc. (Percent)
Build Up	77.95	95.20
Forest	100.00	97.64
Water	100.00	100.00
Bareland	99.22	95.75
Agriculture	98.39	100.00

Figure 8: Accuracy Assessment for LULC 2021

3.5 The Soil and Water Assessment Tool Modelling

SWAT is a publicly available rainfall-runoff hydrology and water quality model. Several categories of data are required as input into the model: Digital Elevation Model, Soil Characteristics, Land Use and Land Cover and Climate Data. DEM is used to delineate and compute the slope of catchments. Soil characteristics are used to determine infiltration and simulate subsurface movement of water. Land Use and Land Cover are used to determine water movement and evapotranspiration. Climate data provides five inputs to the model – precipitation inform of rain or snow, temperature (minimum and maximum), solar radiation, relative humidity and wind velocity. The model simulates vegetation growth, evapotranspiration, hydrology, erosion and sediment movement along with water quality.

Based on the digital elevation model, the model divides a watershed into smaller catchments called sub-basins. The size of sub basins depends on the area threshold, which is set by the user. The model creates smaller spatial units comprising unique combinations of soil, land use and slope, called Hydrological Response Units (HRUs). Based on the sub-basin delineation, SWAT defines streams within each sub-basin called reaches. Although SWAT operates at the spatial unit of an HRU and on a daily time scale. And can be presented in daily, monthly or annual time steps.

The SWAT model was configured and setup for the Thwake River Basin using the input data shown in the Table below.

Table 4: Data Types Used in SWAT and their Sources.

Data Type			Source	Spatial Resolution
Digital Elevation Model (DEM)			United States Geological Survey (USGS)	90m
Land Use/Land Cover Map			Copernicus, Sentinel 2A,2B	20m
Soil map			Food and Agriculture Organization (FAO) Soil database	1:5,000,000
Weather Data			Simulation	Varies (5,500m, 4,000m) Monthly (19,200m, 50,000m)

3.6 Data Preparation for SWAT Model

3.6.1 Land Use and Land Cover

Land Use and Land Cover was obtained from classification of Sentinel 2A,2B imagery of 2020 and preparation of the look up tables required for the data to be usable in the SWAT platform using Notepad. Land Use and Land Cover data was reprojected to the projected coordinate systems (WGS 1984, UTM Zone 37S) and clipped appropriately for use in the Thwake River Basin.

Table 5: Land Use Codes and Percentage of Coverage.

Land Use (Value)	Land Use Symbol (Name)	Percentage Coverage (%)	of Detail
1	FRST	8.01283	Mosaic Forest-shrubland/Grassland
2	FRSE	5.70313	Closed to open broadleaved evergreen/Semi Deciduous
3	URHD	35.3742	Urban High Density and settlements

4	BARR	10.2835	Barren land, Savannah with relatively less bush
5	AGRC	37.6672	Rainfed Cropland
6	WATR	2.7591	Water bodies

3.6.2 Land Topography Data

Land topography in the form of a Digital Elevation Model (DEM) was downloaded from United States Geological Surveys (USGS) Website. The data used was a 90m resolution Digital Elevation Model from Shuttle Rada Topography Mission (DEM SRTM). The SRTM DEM was reprojected and buffered to Thwake River Basin boundary before being used as one of the input files in the SWAT model.

3.6.3 Soil Data

The soil data in form of a digital soil map was acquired from Food and Agriculture Organization (FAO) of the United Nations. The spatial resolution for the soil dataset was 1:5000000, which is equivalent to a pixel size of 500m.

Table 6: Soil Textures and Percentage of Distribution.

Soil Name	Texture	Percentage of Basin
Bc14-2bc-440	Loam	68.1078
Fo48-2ab-42	Sandy-Clay-Loam	0.0127098
Vp45-2-3a-960	Clay	3.16263
Fr7-2a-580	Sand-Clay-Loam	28.7168

Soil Data Coding in Notepad.

Table 7: Soil Data Codes.

Soil Value	Soil Name	Texture
1	Bc14-2bc-440	Loam
2	Fo48-2ab-42	Sandy-Clay-Loam
3	Vp45-2-3a-960	Clay
4	Fr7-2a-580	Sand-Clay-Loam

3.6.4 Climate Data

Climate data was downloaded from Kenya Meteorological Department and Climate Forecast System Reanalysis (CFSR). This website hosts the SWAT – specific weather data which includes: Temperatures (°C), precipitation (mm), wind (m/s), relative humidity (fraction) and solar (MJ/m²). All spatial data were reprojected into a Projected Coordinate System (WGS 1984 UTM Zone 37S). Daily data from 1979 – 2014 was obtained.

3.6.5 Soil and Water Assessment Tool Procedure

ARCSWAT, ArcGIS extension for ArcMap 10.7 Version was downloaded from the SWAT Web portal and installed in readiness for the data analysis. Additionally, MWSWAT and Map Window were downloaded. Map Window in an open-source GIS System contains a plugin for the SWAT2012 database.

Data used in the model are integrated within the GIS interface and are presented in tabular format map information series. The main concepts in the SWAT model processes includes simulation of the hydrological cycle and the water balance leading to the water yield.

The water balance Equation is as follows:

$$SW_t = SW_o + \sum_{i=1}^n (R_{day} - Q_{surf} - E_a - w_{seep} - Q_{gw})$$

Where:

SW_t is Final soil water content (mmH₂O)

SW_o is the Initial water content in a day i (mmH₂O)

t is the Time in days

E_a is the Evaporation on a day i.

R_{day} Amount of rainfall in a day i (mmH₂O)

W_{deep} Amount of water in deep aquifer in a day i

Q_{day} Return flow

Calculation of precipitation based on watershed characteristics.

Where:

$$W_{eff} = \frac{(W - V_i)^2}{(W - V_i - V_{max})}$$

V_i is the Initial abstraction

W_{eff} is the Event flow

3.5.6 SWAT Analyses.

Step 1. ArcSWAT Set Up

New SWAT Project Set Up and Watershed Delineation.

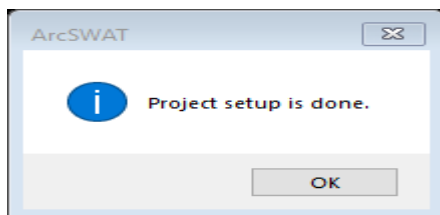


Figure 9: ArcSWAT Set Up.

Step 2. Watershed Delineation.

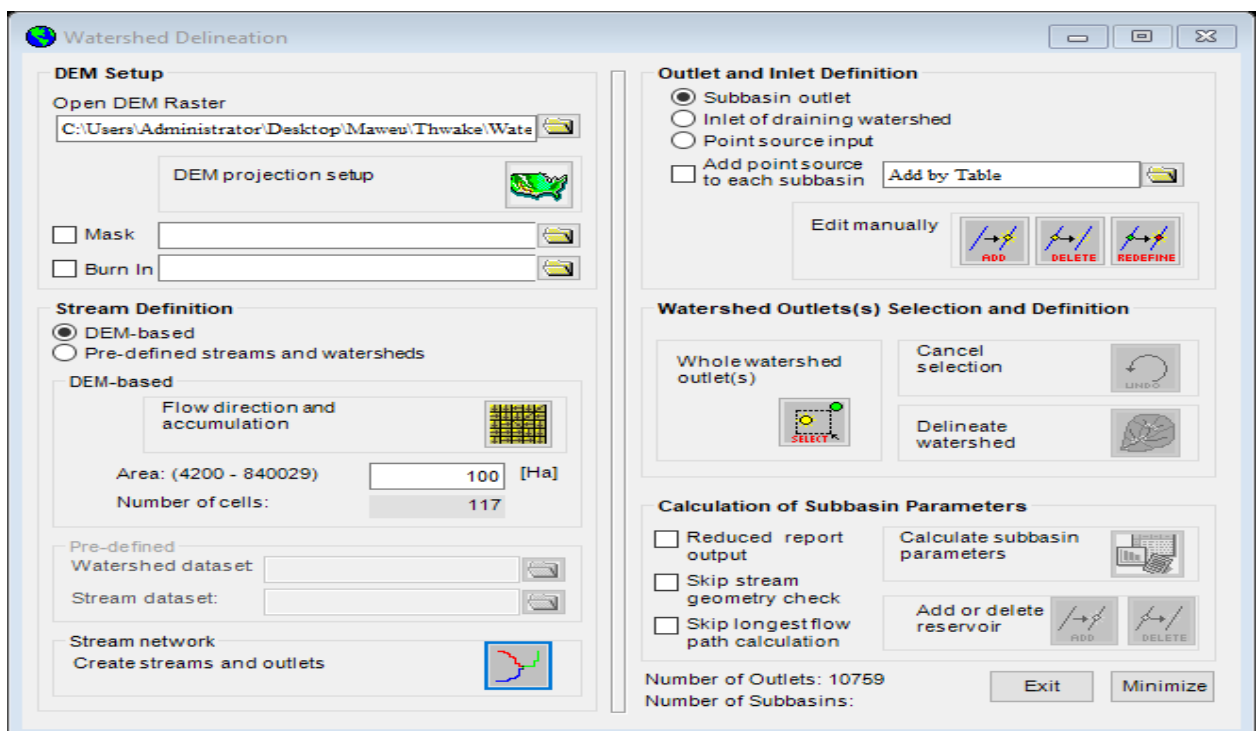


Figure 10: Watershed Delineation.

The SWAT model set up and simulation was a step-by-step process, the set up begun with opening of a new SWAT project in ArcGIS interface. After opening the new SWAT project,

the Digital Elevation Model was loaded to the map document. The DEM was to model water flows downstream by showing the slope, altitude and aspect, this was indicated by the z direction and identification of flow direction and flow accumulation and stream network creation. The area for the flow direction and flow accumulation was altered to 100ha.

This was followed by an automatic watershed delineation to specify the DEM projection set up in order to indicate the z direction. The streams were defined followed by basin boundary delineation and stream network.

The stream network helped in the selection of pour points in order to effectively create streams and outlets. All the sub basins and their outlets were marked by specifying each outlet. Finally, the sub basin parameters were calculated and a topological output report generated to complete the watershed delineation.

Step 3. Land Use, Soils and Slope Definition

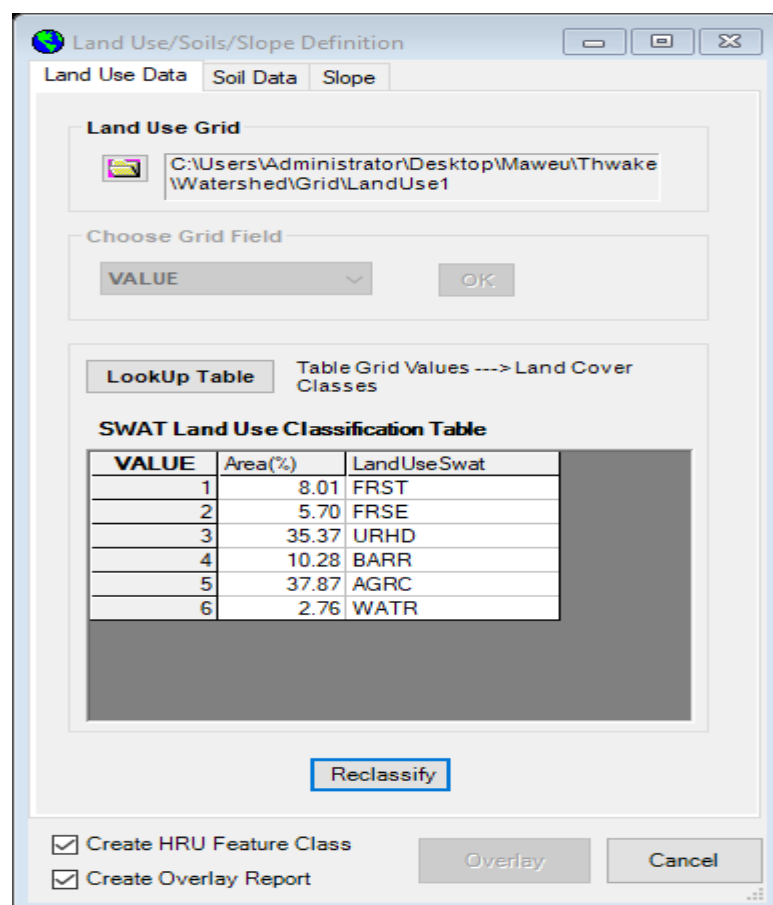


Figure 11: Land use Definition in SWAT.

Land Use/Soils/Slope Definition

Land Use Data Soil Data Slope

Soils Grid

 C:\Users\Administrator\Desktop\Maweu\Thwake Watershed\Grid\LandSoils1

Choose Grid Field

VALUE  

Soil Database Options

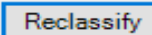
☐ ArcSWAT STATSGO ☐ ArcSWAT SSURGO

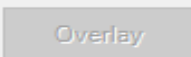
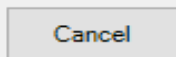
 ☒ UserSoil

LookUp Table Table Grid Values ----> Soils Attributes

SWAT Soil Classification Table

VALUE	Area(%)	Name
1	68.11	Bc14-2bc-440
2	0.01	Fo48-2ab-42
3	3.16	Vp45-2-3a-960
4	28.72	Fr7-2a-580



☒ Create HRU Feature Class  

☒ Create Overlay Report

Figure 12: Soil Data Definition in SWAT.

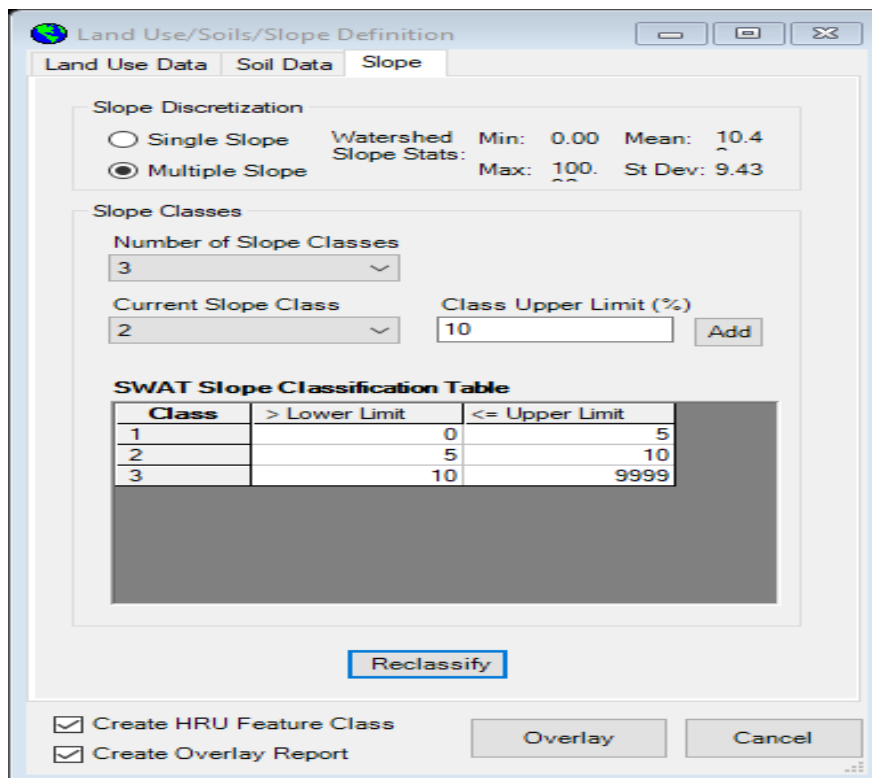


Figure 13: Slope Definition in SWAT.

Land Use and Land Cover data that had been classified and projected was added into the model so that land use and land cover in Thwake River Basin could be reclassified using a look up user table which had been made using Notepad could match with the SWAT land use reclassification. Land use and Land cover was successfully loaded (100 %) and clipped to the watershed boundary.

Soil map that had been downloaded from FAO Website clipped to Thwake River Basin was also added to the model. The projected soil map was reclassified using a look-up user table which had been created using Notepad so that it could match with the soils in the SWAT2012 database table amended earlier. The soil map was successfully loaded (98%), clipped to the watershed boundary and reclassified.

Step 4. HRU Definition.

The study area was also divided into different percentage slope classes ranging from 0 – 5%, 5 – 10% and 10 – 9999%, their total areas in hectares and acres and percentage eater yields per area. The distinctive classes of slope from 0 – 5%, 5 – 10% and 10 – 9999% were necessary for creating Hydrological Recharge Units Feature classes. After successful loading and

clipping of the slope, reclassification was done and an overlay of land uses, soils and the slope was successfully done.

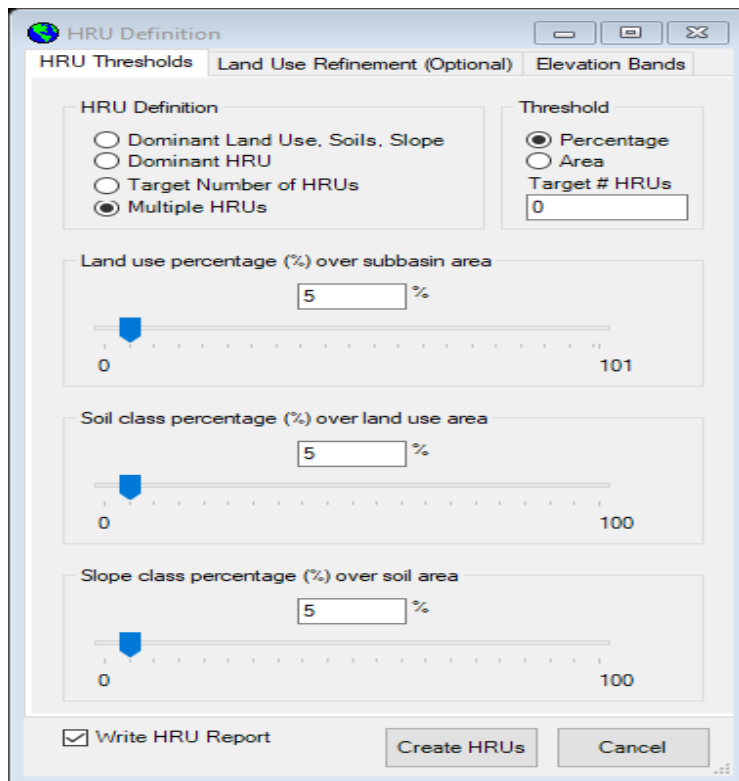


Figure 14: Hydrological Response Units Thresholds.

The watershed was subdivided into Hydrological Response Units, the HRUs exhibit the same land use types, management and soil characteristics. The HRU also increased the model accuracy and prediction of loadings from all the sub-basins. The HRU threshold definition used the combination of land use: soil: slope (5:5:5) percentages. Finally, the HRU report for land use, soils a slope was generated and saved.

Step 5. Weather Data Definition

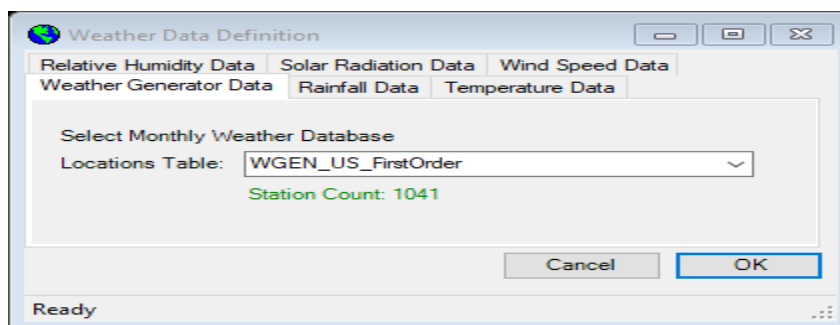


Figure 15: Weather Data Definition in SWAT.

Weather data was generated from the WGEN_US_FirstOrder and simulation of Relative Humidity Data, Solar Radiation Data, Wind Speed Data, Rainfall Data and Temperature Data.

Step 6. Write SWAT Database Tables

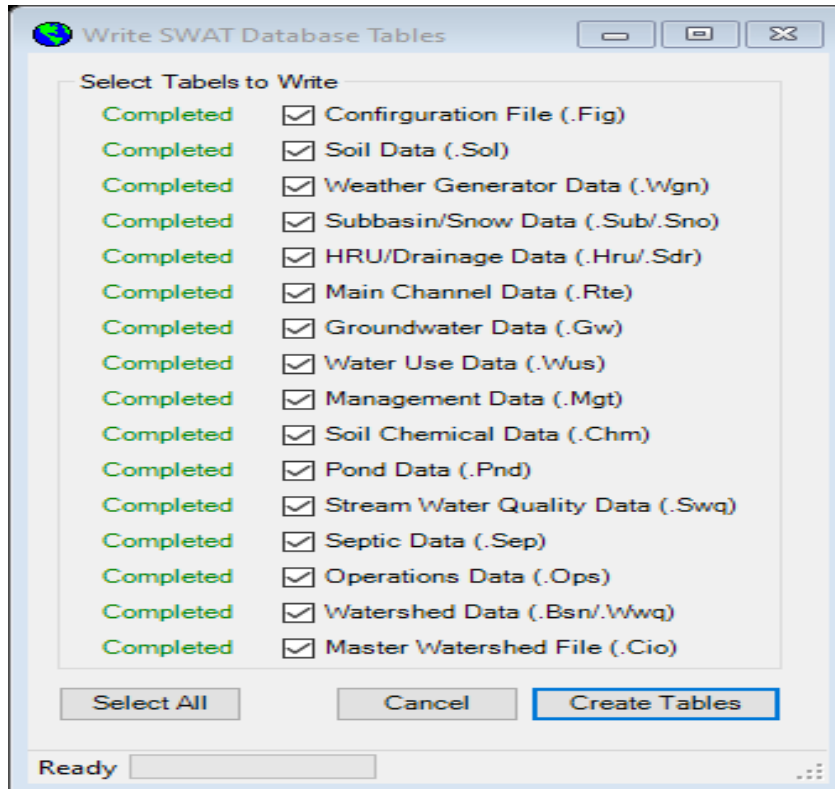


Figure 16: Write SWAT Database Tables.

Different input tables were successfully written including configuration file (.Fig), Soil Data (.sol), Weather Generator Data (.Wgn), Subbasin/Snow Data (.Sun/.Sno), HRU/Drainage Data (.HRU/.Sdr), Main Channel Data (.Rte), Groundwater Data (.Gw), Water Use Data (.Wus), Management Data (.Mgt), Soil Chemical Data (.Chm), Pond Data (.Pnd), Stream Water Quality Data (.Swq), Septic Data (.Sep), Operations Data (.Ops), Watershed Data (.Bsn/.Wwq), Master Watershed File (.Cio). The SWAT input tables were saved in the database ready for analysis.

Step 7. Set Up and Run SWAT Model Simulation

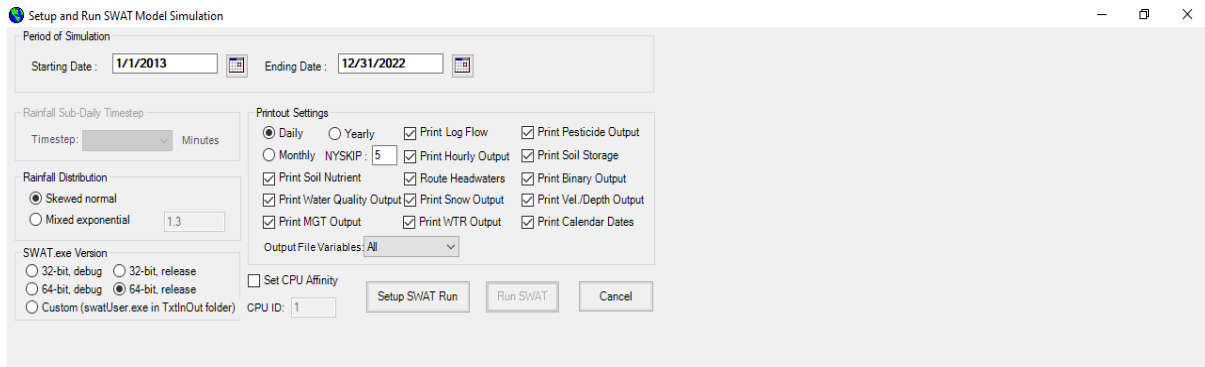


Figure 17: Set Up and Run SWAT Model Simulation.

After a successful model preparation, starting and end period was set in the range with the Hydro-Meteorological data that had been incorporated into the SWAT system in the United States English Format (MM/DD/YYYY) with an NYSKIP period of 5 years to give room for model initial condition effect to check for good performance. The model was then run on a Daily, Monthly and Yearly time step for a period of 10 Years (2013 - 2022)

Step 8. Reading SWAT Outputs.

The databases were imported and an examination of the different files was. The SWAT output definition files were examined to understand the outputs.

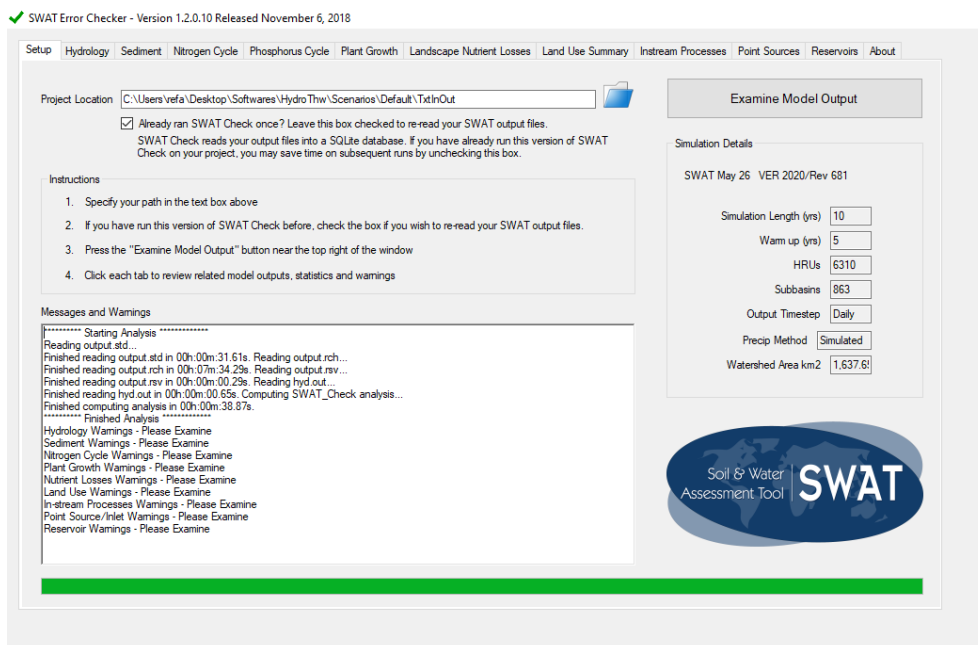


Figure 18: Successful SWAT Run check.

CHAPTER 4: RESULTS AND DISCUSSION

This chapter focuses on the presentation of the overall results produced from the analyses and models that were carried out during the implementation of this project. The overall objective of the study was; To demonstrate the capability of GIS techniques and GIS-based modelling in river basin assessment, management and mapping, in the Thwake River Basin. Consistent with the outlined objectives, the following results were obtained:

1. SWAT Modelling maps.
2. LULC Maps for change detection.
3. LULC Graphs for the change detection trends.
4. SWAT Streams flow in and flow out graphs.

4.1 SWAT Model Map Results.

The SWAT modelling was run successfully for a period of 10 years (2013 - 2022). The inputs to the model were a Digital Elevation Model, Classified Land Use and Land Cover from Sentinel Satellite imagery and Digital Soil Map from FAO. The study area was also divided into different percentage slope classes ranging from 0 – 5%, 5 – 10% and 10 – 9999%, their total areas in hectares and acres and percentage yields per area. The distinctive classes of slope from 0 – 5%, 5 – 10% and 10 – 9999% were necessary for creating Hydrological Recharge Units Feature classes. The HRU threshold definition used the combination of land use: soil: slope (5:5:5) percentages. The soil types present in the river basin are Bc14-2bc-440 (Loam), Fo48-2ab-42 (Sandy-Clay-Loam), Vp45-2-3a-960 (Clay) and, Fr7-2a-580 (Sand-Clay-Loam).

The SWAT simulation's overall results are summarised in the table below.

Table 8: SWAT Run check results

Model Parameters	Output/Results/Features
Simulation Length	10 years
HRUs	5
SUB Basins	6,310
Output Time Step	Daily/Monthly/Annually
Precipitation Method	Simulation
Area/Watershed Area	1637.65 km ²
Warm Up Years	5

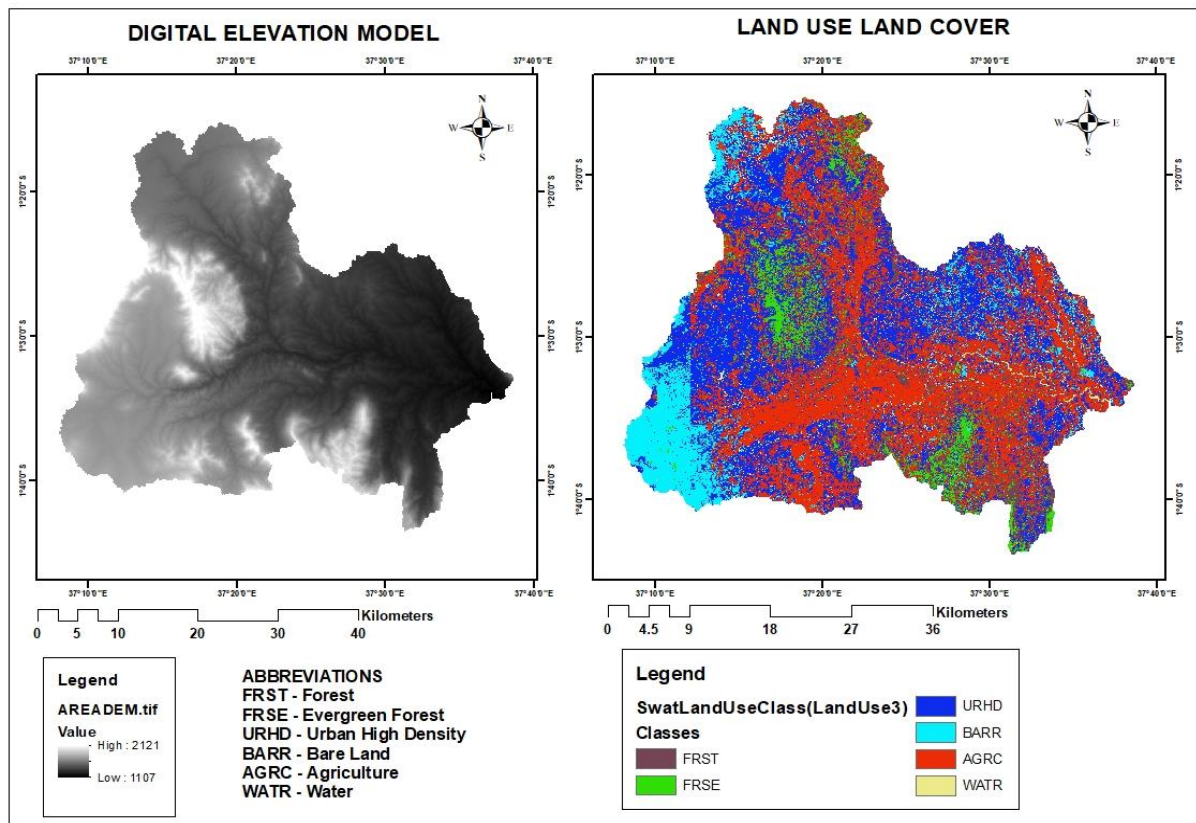


Figure 19: SWAT Model Maps, DEM and LULC

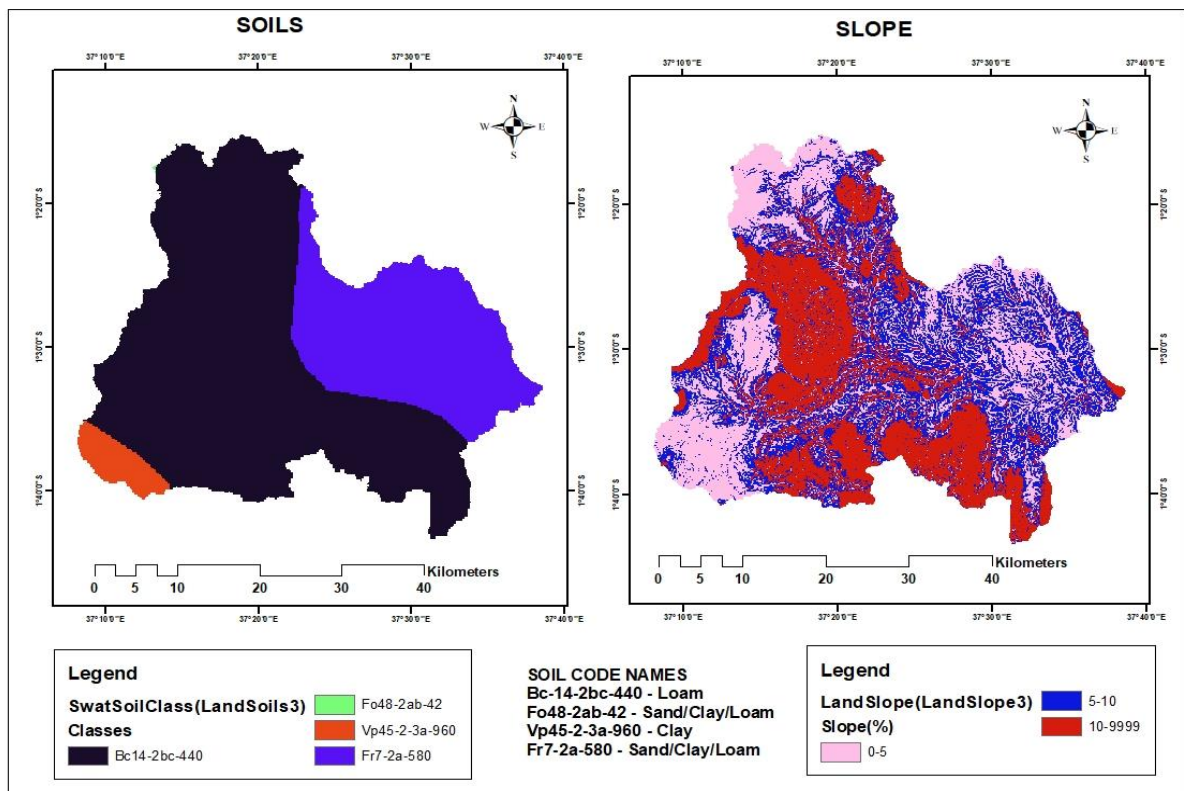


Figure 20: SWAT Model Maps, Soils and Slope Distribution

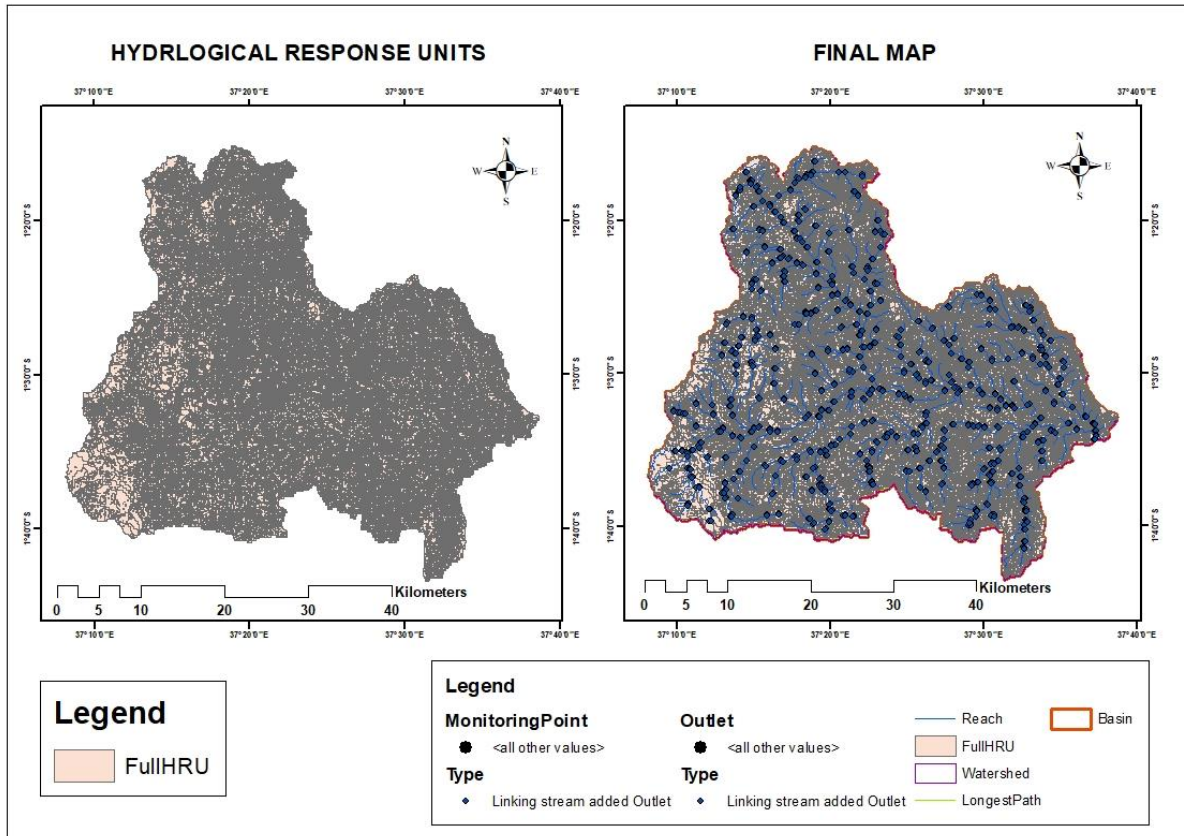


Figure 21: SWAT Model Maps, HRUs and Final Map

4.2 SWAT Stream Flow.

The stream flow data from the SWAT simulation was captured in the imported SWAT2012 database. The stream flow data contains the amount of flow in (cms) and flow out (cms). The number of years skipped from 2013 to 2022 was 5, hence the dates captured were from 2018 to 2022. The amount of flow in and flow out in the streams varied throughout the years with 2019 having the lowest amount of flow in and flow out while 2018 had the highest amount of flow in and flow out. This can be deduced from the fact that; climate change patterns vary at different times of the year. Therefore, the amount of rainfall varies according to Figure 2:

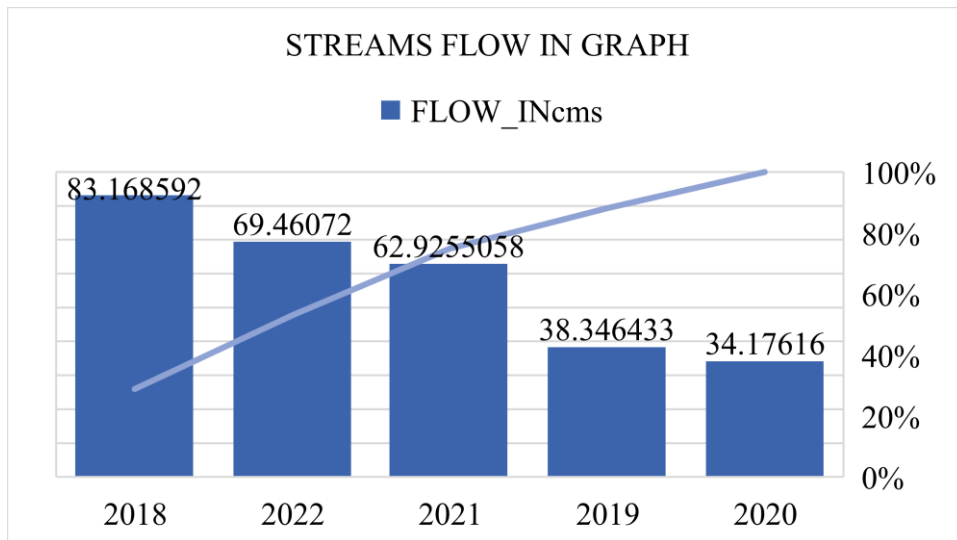


Figure 22: SWAT Model Streams Flow INcms.

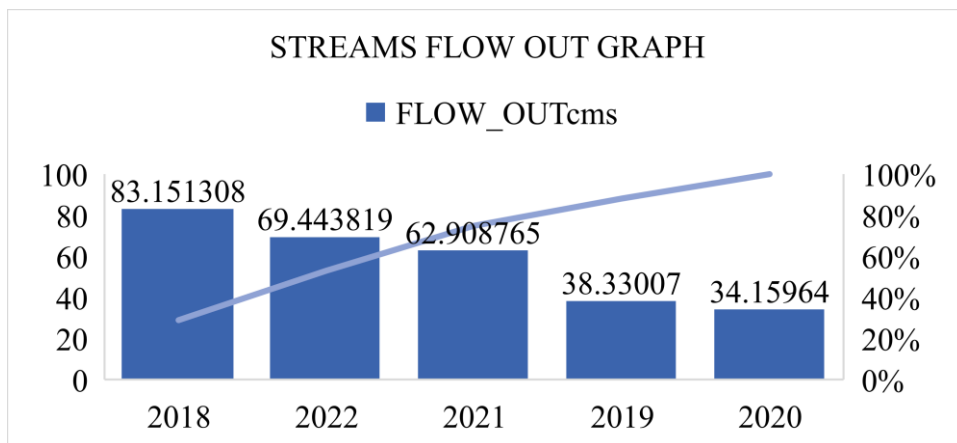


Figure 23: SWAT Model Streams Flow OUTcms

4.3 LULC Change Detection Maps

Satellite imageries for the years 2013, 2017 and 2022 were classified and the trends for change detection are shown in the Figure 24: LULC Trends. The satellite imageries were classified into 5 classes each which are Vegetation, Bare land, Build Up, Water and Agriculture. The classes can be densified according to Table 3: Classification Scheme for the classification and Change Detection Procedure.

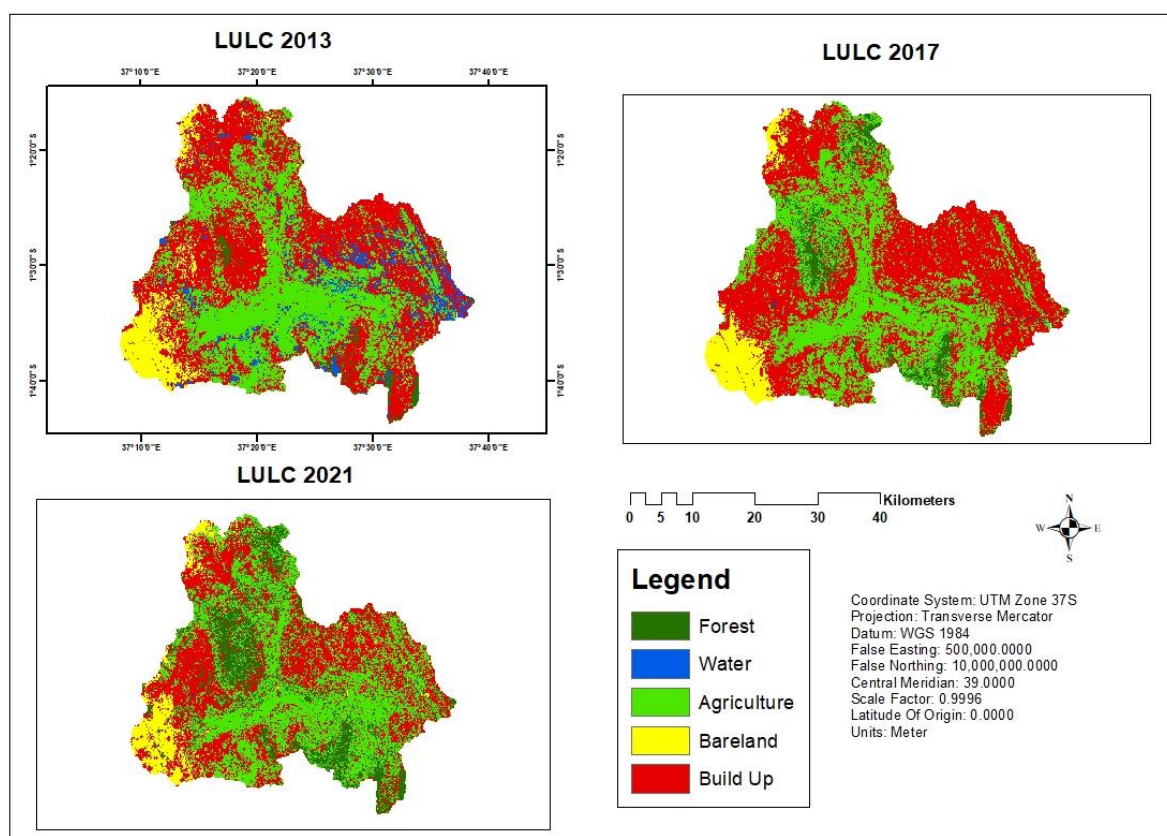


Figure 24: LULC Trends

With the accuracies provided in Figure 6: Accuracy Assessment for LULC 2013., Figure 7: Accuracy Assessment for LULC 2017. And Figure 8: Accuracy Assessment for LULC 2021. The trends in the LULC classifications were plotted in the graphs below.

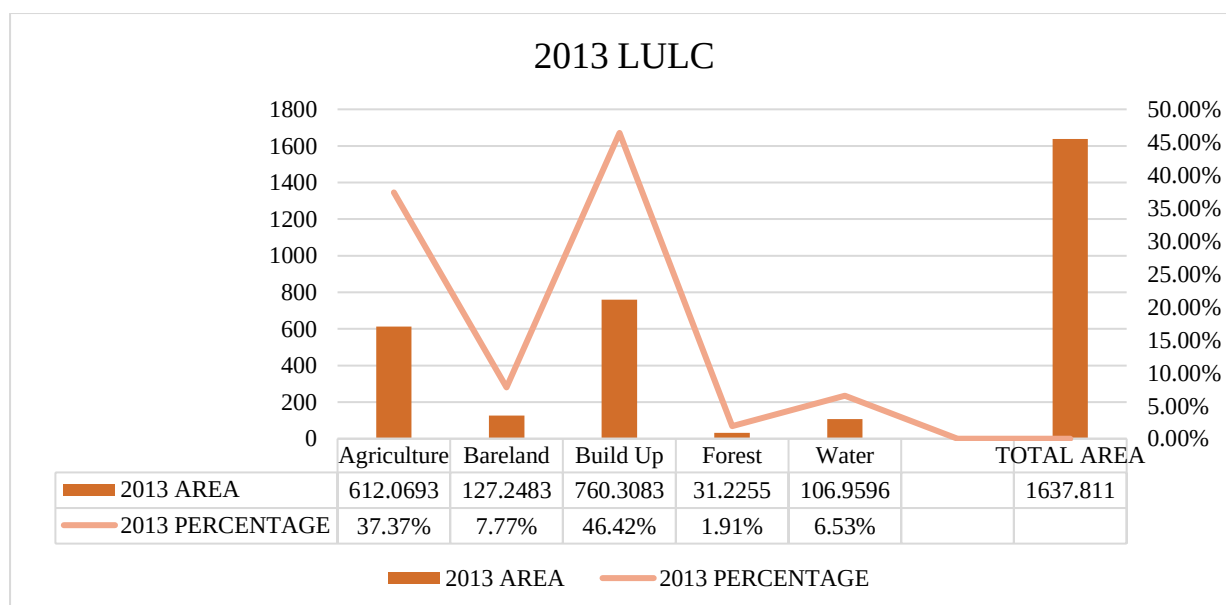


Figure 25: LULC Graph for 2013.

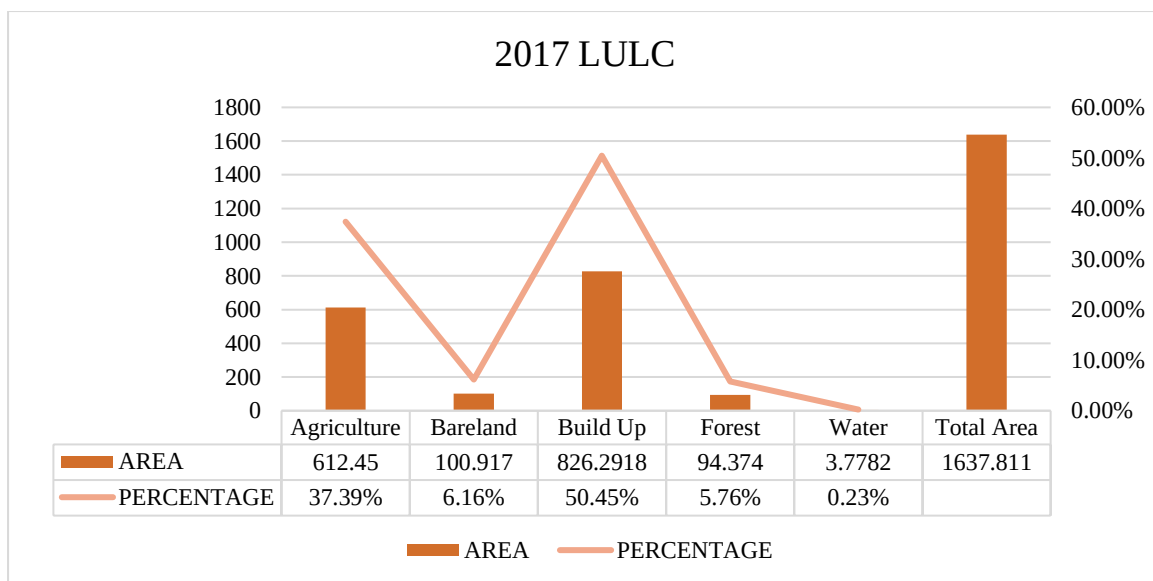


Figure 26: LULC graph for 2017.

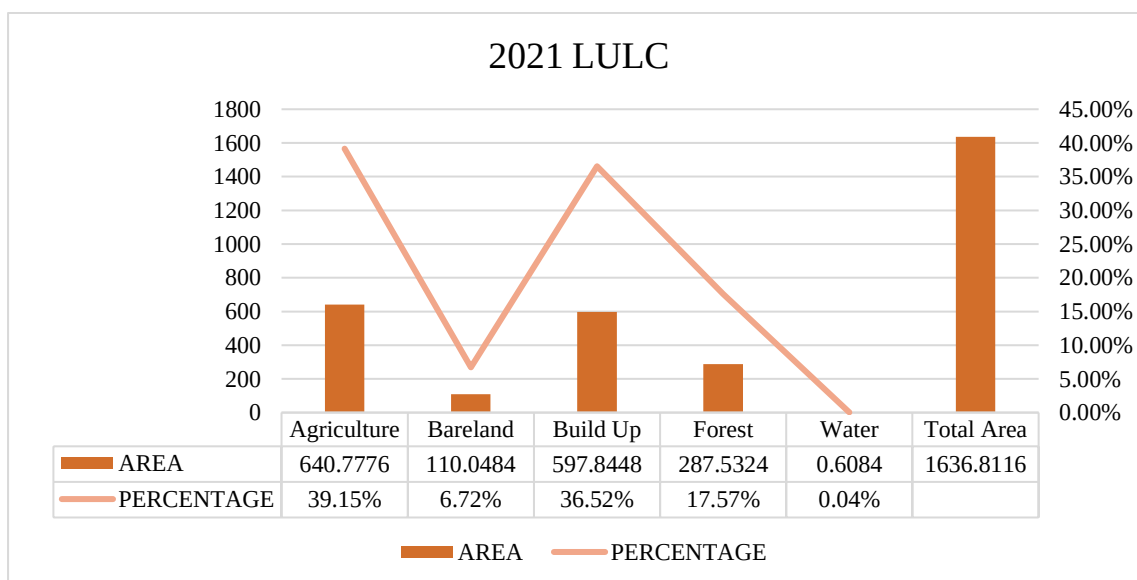


Figure 27: LULC graph for 2021.

The positive changes were found to be in agriculture, bare land and forest. The negative changes were found in Water and Build Up. The growth of Agriculture was found to be along river basin valleys and the bare land was found to provide space for build-up and forest. The amount of forest cover was noted to be along the high-altitude regions in the river basin according to Figure 24: LULC Trends. The Water class was found to be decreasing with increase in agriculture along river basin valleys. These positive and negative changes for the years 2013, 2017 and 2021 can be summarised in the graph below.

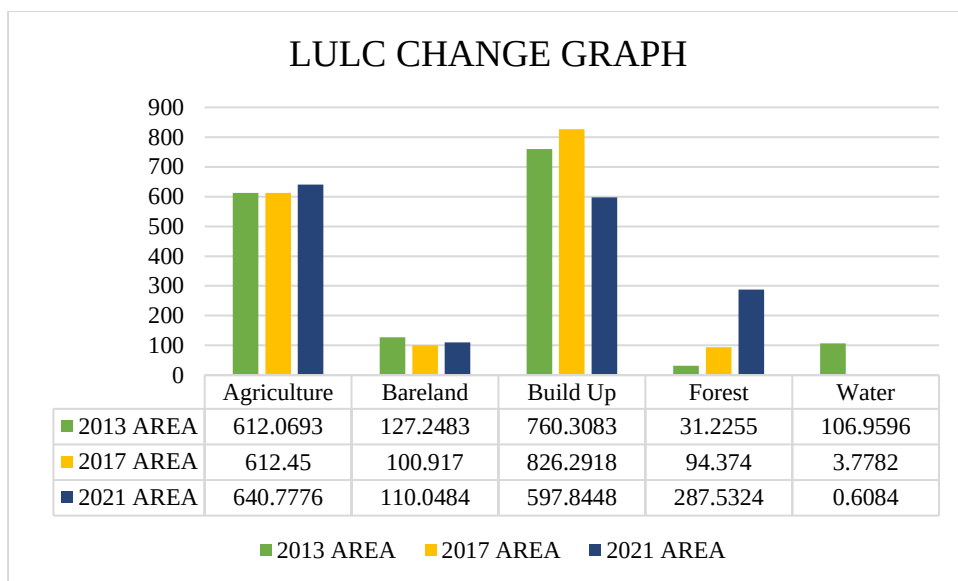


Figure 28: Overall LULC Graph.

Table 9: Positive and Negative Impacts

Category	2013 Area (km ²)	2013 % Cover	2017 Area (km ²)	2017 % Cover	2021 Area (km ²)	2021 % Cover	Impact (Positive/Negative)
Agriculture	612.0693	37.37	612.45	37.39	640.7776	39.15	Positive
Bare land	127.2483	7.77	100.917	6.16	110.0484	6.72	Negative
Build Up	760.3083	46.42	826.2918	50.54	597.8448	36.52	Negative
Forest	31.2255	1.91	9.374	5.76	287.5324	17.57	Positive
Water	106.9596	6.53	3.7782	0.23	0.6084	0.04	Negative
Total	1637.811	100	1637.811	100	1637.811	100	

CHAPTER 5: CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

In this chapter, the conclusions are derived from the findings of this research on assessing the impact of Land Use and Land Cover changes on hydrological processes in the Thwake River basin using GIS-Based models and remote sensing techniques. The conclusions are based on the objectives, research questions and purpose of the research. The implications of the results and the resultant recommendations are also explained. The recommendations were based on the conclusions and purpose of the research.

5.2 Conclusions

Using the analysed results, the following conclusions were reached:

Land Use and Land Cover can have a significant impact on the hydrology of a river basin, which in turn affects the flow regime. This is evident in the Thwake River basin where variability in annual rainfall has caused irregular stream flow levels, influencing the availability and distribution of water in different watersheds and sub-basin. Prolonged dry periods also reduce the water table, causing streams to dry up.

The Soil and Water Assessment Tool (SWAT) has been found to be an effective model for simulating the hydrological conditions in the Thwake River Basin. The model involves the use of datasets such as a Digital Elevation Model, a Digital Soil Map, and a classified Land Use and Land Cover Map. The weather data is also defined and simulated by the model. Although the outcome of this research has limitations, it can still be used in the management of water resources in the Thwake River basin and neighbouring regions due to the existence of a spatial-temporal relationship between rainfall and river flow.

The impacts of Land Use and Land Cover changes on the hydrology and flow regime of a river basin are significant and require proper understanding and management. The SWAT model provides valuable insights for effective water resource management in the Thwake River basin and similar regions facing similar challenges.

5.3 Recommendations

Given the above conclusions, the following recommendations were done:

One of the key recommendations for river management organizations and institutions is to put in place a water and climate variability framework. This framework should aim to manage and

sustain the available water resources in the Thwake River basin, taking into account the impact of changes in Land Use and Land Cover on hydrological processes. The framework should consider the interplay between the different water users and the environment, and ensure that the water resources are used in an equitable and sustainable manner.

Another recommendation is to improve the technologies for managing water resources. This can be achieved by using more accurate GIS models and applications, which can help monitor the quality of water resources in the Thwake River basin. This will allow for a better understanding of the impact of Land Use and Land Cover changes on the hydrological processes and the availability of water resources. In addition, the use of remote sensing techniques can help provide high-resolution data on the spatial and temporal distribution of water resources.

The study recommends more research in the area of water management in the Thwake River basin, employing other spatial techniques such as modelling the socio-economic environmental concerns within the basin. Hydrological modelling should be encouraged, even at a micro-scale, to better understand the impact of Land Use and Land Cover changes on the water cycle in the basin.

Finally, more consideration should be made to the availability of data, data processing, retrieval and storage to avoid errors. Accurate data is essential for effective water management and the assessment of the impact of Land Use and Land Cover changes on hydrological processes.

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